Supplementary Information

Harvesting names from unlabeled papers to augment materials named entity recognition on unseen documents

Hatem Madi

Supplementary Notes 1 to 13 give the protocol record, the compute and repeats, the corpora's composition, and the analyses the main text summarizes; Tables S1 to S16 are numbered in order of appearance. Table and figure numbers without an S refer to the main text, whose sections are cited by name; the methods are all in the main text's Methods.

Supplementary Note 1: the protocol record

The repository names its protocols by the stage of the project in which they were fixed, a "week"; Table S1's first column gives that name with each protocol's decisions, and the main text cites the protocols by date. The week-12 screen's variants are labeled V1 to V8, as Supplementary Note 7 lists them. The last rows give the companion study's protocols whose results the main text or these Notes quote; their decisions are not among the twelve of Table 2 or the twenty counted in the main text (Results, *What else was tried, and the count*).

Table S1. The protocol record of the names route, then of the companion study's results quoted in the main text or these Notes: when each protocol was committed (commit times, UTC; the protocol text's own written times are given below the table), when its runs ran, when its result was recorded, and the amendments.

Decision	Protocol committed (times UTC)	Runs	Result recorded	Amendments
the earlier protocols (Results, *What else was tried, and the count*): mention replacement on MuLMS (week 10); names from the other subfields' sentences (week 11)	17 October 20:21 (`a08356f`); 18 October 01:13 (`427e34c`)	17–20 October	week 10: 18 October 00:48 (`c2ca3f6`); week 11: 18 October (MuLMS, `14ed50f`) and 20 October (MaterioMiner, `04764ef`)	—
D1, D2 (week 12)	18 October 11:00 (`40631ce`; refined 11:05, `bd20cec`): the screen on seeds 0–200, the decisions on seeds 300–900 so that the selection's seed noise does not enter them; the MaterioMiner analogue gated on D1; the methods criterion: D1 plus a MaterioMiner pass	20–24 October	24 October	three amendments before any run added variants V4, V6, V7–V8 to the screen
D3, D4 (week 12)	18 October 11:00 (`40631ce`): the winner's MaterioMiner analogue decided against the CRF head per granularity, to run only if D1 passed; 20 October 00:51 (`ce0f592`): the analogue's batches, with its comparisons against MR3 and the votes pre-specified; the gate lifted at 09:14 (`808ff4b`), restored at 13:53 (`5f7f827`), lifted again at 18:42 (`de62446`)	24–25 October	25 October (`3432988`)	—
D5, with 13.1a, 13.1b, 13.1d and 13.3a, 13.3b (week 13)	20 October 09:14 (`808ff4b`): the pooled decision over both corpora as a second route to the claim, "each unit its 10-seed mean", the stop rule lifted; amended three times the same day, before any decision run of the route: 09:25 (`e94f352`), the label-propagation rule's two MaterioMiner decisions and an SOFC-Exp comparison pre-specified after its MuLMS results (Supplementary Note 7); 13:53 (`5f7f827`), eighteen minutes after the screen chose MR6 + names, D6 added conditional on D1 and the analogue, which the morning's text ran whenever the screen had a winner, gated on D1; 18:42 (`de62446`), D6's condition widened to D1 or D5 and the analogue's morning condition restored, the gating recorded as a mistake	24–25 October (13.1a, 13.1b and 13.1d on saved predictions)	D5: 25 October 01:11 (`3432988`); 13.1a, 13.1b: 20 October 09:25 (`e94f352`); 13.1d: 09:47 (`aca3a4f`); 13.3a: 24 October 17:26 (`2432c8c`); 13.3b: 25 October 02:45 (`6fb8ddd`)	the analysis took MuLMS's 10-seed means; the 7-seed reading is D5′

Decision	Protocol committed (times UTC)	Runs	Result recorded	Amendments
D6 (week 13)	20 October 13:53 (5f7f827), conditional on D1; widened to D1 or D5 at 18:42 (de62446)	25 October	25 October	—
D7, D8 (week 14)	29 October 12:07 (2886267), two minutes after the first batch was pushed and five hours before its first result, after MR6 + names' own results on the same units	29 October	29 October	—
ablations (week 14)	29 October, as above	31 October	31 October	—
the lexicon-membership analysis (week 15; post hoc, no decision)	31 October 16:13 (2c60f33): the rebuild, its checks and the reading fixed before the run	31 October (w15a)	31 October	—
D9, D10 (week 17; follow-up)	31 October 19:15 (f39447c), after D1 to D8 had been written up (the lexicon split's results arrived 26 minutes later), a pass expected to be unlikely; the code 19:41 (013f9ac)	31 October 19:58 to 1 November 00:35 (the first jobs started at 19:58, w17b, and 19:59, w17a; w17b ended at 23:00)	1 November	—
D11, D12 (week 18; follow-up)	1 November 10:51 (e02c3be), at the request of an internal review of the draft (main text, Methods, *Use of an AI assistant*), after D1 to D10 and the lexicon split were known; the names' side the stored decision runs of 24 October, only the control new; the code 10:58 (7906996) and the analysis 11:04 (789f42e), both before any run	1 November 11:14 to 13:43 (w18a, pushed at 11:12, pulled at 13:58)	1 November 14:06 (13506a4); the post hoc nearest-recall reading added after it, 14:16 (ea8a92e)	—
companion, week 5: multi-task training against the CRF head on MaterioMiner's held-out papers, pre-specified without decision (fine-grained +3.72 [+0.57, +6.87]; multi-task training's own decision was on the random splits); and self-training on the fatigue corpus against the CRF head there, a decision that failed (+1.74 [−1.16, +4.64] fine-grained, +1.56 [−0.88, +4.00] coarse-grained; its votes too), which the names route reports again as V6's analogue (Supplementary Note 12)	15 October 15:50 (6f2dc11)	15 October	15 October 23:20 (1b30d82)	—
companion, week 7 part 1: label propagation within a document, six decisions on saved predictions	16 October 11:46 (cc73d37)	saved predictions	16 October 11:53 (MaterioMiner, 93966bb) and 13:54 (SOFC-Exp, ceabb78)	11:57 (e41aa44): retyping made a decision on SOFC-Exp, after MaterioMiner's results and before SOFC-Exp's
companion, week 7 part 2a: recall calibration, a bias on the CRF head's emissions chosen on validation folds; five decisions, none passing (fine-grained −0.59 [−1.75, +0.57], coarse-grained +0.17 [−0.14, +0.48], SOFC-Exp −0.21 [−0.58, +0.17] paired by seed)	16 October 13:11 (61c9520)	16 October (w7a)	16 October 15:33 (503f690)	—
companion, week 7 part 2b: MR3 against the CRF head on MaterioMiner's held-out papers, both passing (+2.77 [+1.51, +4.03] fine-grained, +1.28 [+0.06, +2.51] coarse-grained), and on SOFC-Exp's official split, failing (−0.29 [−1.04, +0.47] paired by seed)	16 October 14:39 (494cac1): a 3-seed screen, then 10 seeds	17 October (w7c 03:42–04:15, w7d 04:19–06:47)	17 October 06:59 (db7c78d; the screen at 04:19, 7d55340)	—
companion, week 8 part 1: SOFC-Exp's random sentence splits and article folds, the CRF head's 86.27 and 82.79 of the Introduction (pre-specified without decision)	16 October 17:24 (c06fc5e)	17 October	17 October 10:11 (8cd8fba)	—

Decision	Protocol committed (times UTC)	Runs	Result recorded	Amendments
companion, week 9: multi-task training with MR3 against the CRF head, pre-specified without decision (fine-grained +6.73 [+6.03, +7.43]; its single-model decisions on MaterioMiner were against the multi-task model); MR3 on a MaterioMiner-sized SOFC-Exp, failing (+0.50 [−0.47, +1.46])	17 October 07:28 (`536cbe7`)	17 October	17 October 18:50 (`81134b7`) and 19:30 (`65c17b9`)	12:30 (`fa69d9a`): part 1 runs from part 2's bundle, before any result
companion: seen and unseen test names and the decomposition of the recall gap (post hoc, on saved predictions; no protocol)	—	saved predictions	MaterioMiner 16 October 17:00 (`3c06c11`); MuLMS and SOFC-Exp added with their runs	—

Times. Protocol and result times are the commits' (UTC); push and pull times are the session notes'; a session's end or length in Supplementary Note 11, and weeks 17 and 18's run times, are those its Kaggle jobs wrote in their manifest (each job's start plus its minutes, by the machine's clock), other run times the session notes'; and GitHub's records date only the pushes of commits. When the pooled decision was added at 09:14 on 20 October, the screen's first half was recorded (MR6 and the gazetteer variants, at 08:11) and its second half, with MR6 + names, was not (13:35); the 18:42 amendment followed the spread between subfields that showed the 7-subfield decision underpowered. The protocol text's own written times differ from the commits': later by 6 to 53 minutes in the 09:25, 09:47 and 13:53 commits of 20 October (which carry 09:50 and 09:55, 10:40, and 14:00; the 09:47 commit recorded the label-propagation rule's results on SOFC-Exp's folds and MaterioMiner), earlier by about four hours in the 18:42 one (14:30), and earlier by about six minutes in the week-17 protocol (written 19:10, committed 19:15:57). Week 16 was the companion paper's write-up and has no protocol.

Push times. Two records kept by GitHub, archived in the repository (`protocol_pushes.py`), date each push independently of the commit times: the public event feed (`results/github_push_events.tsv`) and the repository's activity log (`results/github_push_activity.tsv`), which lists every push to a branch with the commits before and after it, except merges made through pull requests. By the activity log every protocol commit of Table S1, from the companion study's of 15 October to the follow-up's of 1 November, reached GitHub 1 to 7 seconds after its commit time (for example `40631ce` at 11:00:59 on 18 October, `808ff4b` at 09:14:24 and `de62446` at 18:42:57 on 20 October, `2886267` at 12:07:40 on 29 October, `f39447c` at 19:15:59 on 31 October and `e02c3be` at 10:51:13 on 1 November). The event feed lacks some pushes and held none after 30 October 22:51 when last archived; on every push both records hold it agrees with the activity log within two seconds, and within one on every protocol's push. The activity log's times for the pushes of 1 November between 08:00 and 09:00 UTC, none of them a protocol's, read one hour late (the hour that US Pacific time repeats when daylight saving ends that day): the commits they carried had been built on by commits pushed before the times the log gives them.

Supplementary Note 2: compute and repeats

The training settings are in the Methods (*Systems*). On MuLMS's decision seeds a CRF head run takes 96 s on a T4 (the best epoch after 8.1 epochs on average), an MR3 run 198 s (5.7) and an MR6 run 296 s (5.1). A names run's stored time is its job's: the three teachers, their tagging of the corpus and the job's three or four students together, 2,099 s on average over the 14 jobs (1,598–2,972 s; the best epoch after 5.6 epochs). Per student that is 609 s (the job's time over its students, averaged over the 14 jobs), 6.4 times the CRF head's 95.8 s; the harvest alone, the job less its students' own training at MR6's rate, is about 1,063 s, eleven CRF-head runs, and a single student charged with the whole harvest costs about 14 times the CRF head's run. None of these figures counts building the corpus (the search, download and filtering of about 300 articles). The students see 2.8 to 4.6 times more sentences per epoch (601, 1,203 and 1,209 augmented sentences on average beside the 340 gold ones, for MR3, MR6 and the names) and reach their best epoch after 5.7, 5.1 and 5.6 epochs against the CRF head's 8.1, so up to it they take 2.0, 2.9 and 3.1 times its updates. Runs used Kaggle sessions of two T4 GPUs with deterministic CUDA, one frozen commit per batch, a pinned image and pinned libraries; the three week-14 batches alone took 19 session hours (5.2, 4.3 and 9.7). The two week-17 sessions took 4.7 and 3.1 hours. A names job on the multi-task base (three multi-task teachers, their tagging of the corpus with two heads, ten students) took 82.5 to 122.7 minutes (mean 99.7, by the results files' run times; the session manifests give 82.7 to 122.9, mean 99.9), 10.0 minutes per student with the harvest shared over its ten students; the harvest alone, the job less ten students at the control's rate, is about 31 minutes, so a student charged with all of it costs about 38. A multi-task six-copy control run takes 6.9 minutes, a three-copy one 4.8, the multi-task model 2.0, and a CRF-head run 1.1 fine-grained and 1.3

coarse-grained on the same papers; the students' best epoch came after 5.4 to 8.1 epochs on average per paper and granularity.

Every interval in the paper is the analysis scripts' value, computed from per-unit values (F1 means over seeds, or rates pooled over seeds) rounded to two decimals; recomputed from the unrounded runs, no bound moves by more than about 0.01 and no verdict changes.

Repeats. From week 5 on (15 October, 16:00 UTC), every GPU session that completed, except two, began by repeating an earlier run, and every repeat was identical (the sessions that failed or hung left no record of their jobs: five on 17 October, before week 10, whose bundle had a new version within about 20 minutes of their push, and two of weeks 10 to 18, items 4 and 5 of Supplementary Note 11): one batch of 15 October (`w5a`, run 16:05 to 22:10 UTC) ran none, and the second session of the follow-up of 31 October (19:15) (`w17b`) instead reproduced the stored multi-task runs' scores through its teachers. The week-4 sessions of 15 October, which wrote the CRF head's first GPU reference runs, began without a repeat after a smoke session that ran one CRF run three times, with identical scores and predictions.

Supplementary Note 3: the corpora

The labeled corpora. MaterioMiner's papers are DOIs 10.1007/s11340-021-00758-x (s11340), 10.1038/ncomms7164, 10.1038/s41467-020-19071-7 (s41467) and 10.3390/met10111458. Its fine-grained classes are chosen as the Methods (*Data*) give. MuLMS's kept entities by type: MAT 12,647, VALUE 4,628, PROPERTY 3,880, FORM 3,180, MEASUREMENT 2,106, CITE 1,704, SAMPLE 1,391, TECHNIQUE 920, DEV 757, INSTRUMENT 370, NUM 260, UNIT 91, RANGE 18.

The unlabeled corpora. Their construction is in the Methods (*Systems*). Per corpus: 290–368 articles with text (292–370 selected), published between 1996 and 2026 (over 99% from 2009 on), 63–90% of them CC BY and 10–33% CC BY-NC or ND (0.3–4% without a licence field), the largest journal holding 13–63% of the articles (MDPI Materials 52% of the steel corpus and 63% of the fatigue corpus). Each report records the query, the counts at every stage, the article identifiers with licences and the corpus's SHA-256; a rebuild that differs is detected by the hash.

Supplementary Note 4: the lexicon's settings

Settings (pre-specified, 3 seeds). Each ablation (Table S2) changes one setting and runs on the 7 subfields at the screen's seeds (0 to 200) against the runs of MR6 + names on those seeds, which are the selected variant's and so a slightly flattering reference.

Table S2. The lexicon's settings (pre-specified, 3 seeds): distinct names, the share of the unseen test names in the lexicon (by the runs' stored counts, which call a test name unseen when no training sentence annotates its words with its type; for MR6 + names this gives the 22.2% of the main text's rule (Methods, *Data*) too), micro F1 on the 7 subfields and the differences from MR6 + names, MR3 and the CRF head. A corpus's "first" 10,000 or 20,000 sentences are the first of one fixed seeded shuffle of its 40,000, a random sample (Methods, *Systems*).

Setting (the value in MR6 + names)	Distinct names (summed over the 7 subfields)	Unseen test names in the lexicon	Micro F1	Δ vs MR6 + names [95% CI]	Δ vs MR3	Δ vs CRF head
MR6 + names: 40,000 sentences, threshold 0.9, 3 of 3 teachers, a uniform pool	54,095	22.2%	52.09		+1.45 [−1.35, +4.25]	+3.16 [−1.96, +8.28]
corpus: its first 10,000 sentences	19,171	16.4%	51.94	−0.15 [−1.78, +1.48]	+1.30 [−1.14, +3.74]	+3.01 [−1.21, +7.23]
corpus: its first 20,000 sentences	32,461	20.1%	51.93	−0.16 [−2.45, +2.14]	+1.30 [−0.56, +3.15]	+3.00 [−0.49, +6.49]
filter: marginal threshold 0.7	77,879	31.5%	51.93	−0.16 [−1.62, +1.31]	+1.29 [−0.95, +3.53]	+3.00 [−1.85, +7.85]
filter: 2 of 3 teachers agree	78,604	28.8%	52.20	+0.11 [−1.58, +1.81]	+1.56 [+0.07, +3.05]	+3.27 [−0.85, +7.38]
pool: harvested names in half of the draws (its uniform pool draws one in at least 92%)	54,095	22.2%	52.18	+0.09 [−1.63, +1.80]	+1.54 [−0.74, +3.82]	+3.24 [−0.73, +7.22]

Summed over the seven subfields, the lexicons range from 19 thousand to 79 thousand distinct names and from 16% to 32% of the unseen test names, and micro F1 stays within 0.16 of the 52.09 of MR6 + names; every interval against it covers 0, and per subfield the differences run from −3.2 to +2.9, the spread of 3-seed means. These readings detect differences of 2 to 3 points: every interval lies within −2.45 and +2.14, so larger effects are excluded and smaller ones are not; the settings are not shown to be equal. Within that limit, a quarter of the corpus (a quarter of the teachers' tagging) with the strictest filter is not distinguishable from the rest, consistent with the main text (Results, *Whose gain is it*): if the names add under a point, the settings that govern how many names enter cannot show an effect. On the unseen names the settings do move the trade of the main text (Results, *What the names do*) (post hoc; the screen's 3 seeds pooled over the subfields, against the screen runs of MR6 + names at 48.1% recall and 50.0 precision, computed from their saved predictions): smaller lexicons recall fewer at higher precision (10,000 sentences 46.9% at 52.3; 20,000 46.2% at 53.0; the names in half of the draws 46.2% at 53.7), a lower threshold more at lower precision (0.7: 48.8% at 49.5), and 2 of 3 teachers a little more of both (48.3% at 50.6), while F1 on the unseen names stays between 49.1 and 49.7.

Supplementary Note 5: what the names do, in detail

Everything in this Note is post hoc, except the companion study's decisions quoted at its end.

Rates. Over all ten seeds the names system makes, net of the predictions it drops, 474 more unseen-name predictions per run than MR6 and 143 more correct ones, a net rate of 30% (MR3's net additions over the CRF head, 43%; MR6's over MR3, 33%); these are differences of pooled counts, not the precision of the predictions one system makes and the other does not. On the decision seeds alone the pooled trade against MR6 is 3.0 points of recall for 2.9 of precision and +0.5 F1 on unseen names (Table 4's note). The mixed corpus's figures are similar (46.7% recall at 52.4 precision, against the topic corpus's 46.9% at 51.3 on the same seeds), and its unseen-name F1 is slightly higher (49.35 against 49.0). On MaterioMiner's papers, fine-grained, the names add to both pooled rates on unseen names (39.7% to 41.1% recall and 49.2 to 53.2 precision over MR6; F1 43.9 to 46.4), and coarse-grained a little of each (53.0% to 54.2% and 52.4 to 53.2; F1 52.7 to 53.7). Paired over the four papers the changes are +0.86 [−3.38, +5.10] in recall and +1.32 [−6.62, +9.25] in precision fine-grained, and +1.60 [−1.37, +4.57] and +1.16 [−2.24, +4.57] coarse-grained, every interval covering 0.

Per subfield, the lexicon's coverage of the unseen test names runs from 7% (semiconductors) to 36% (electrolysis), and the names' gain over MR6 on the decision seeds from −2.3 (polymers) to +3.3 (solid oxide fuel cells); the two correlate negatively (Pearson r = −0.71, Spearman −0.61, 7 points), and the gain also correlates with how weak the CRF head is on the subfield (r = −0.76 with its F1; semiconductors at 32.5 and solid oxide fuel cells at 43.5 gain most). The two readings cannot be separated here. Coverage and the baseline's F1 are themselves correlated (r = 0.89); given F1, coverage keeps almost no association with the gain (partial r = −0.10) while F1 given coverage keeps some (−0.40), but with 7 points, 4 degrees of freedom for a partial correlation, neither is distinguishable from 0, and the raw correlations fall to −0.19 and −0.01 without the two subfields that drive them (semiconductors and solid oxide fuel cells). The raw correlation with coverage runs against a simple dose–response in F1, though the membership split (main text, Results, *What the names do*) finds the recall gain on the held names in every subfield; F1 also pays the precision the names cost.

The lexicon split in full. Table S3 gives the levels and every paired comparison of the analysis in the main text (Results, *What the names do*), and Supplementary Note 13 each unit's gains and the unseen-name trade of every system against the one it builds on, in F1, recall and precision. A test name is unseen by the main text's words-only rule (Methods, *Data*) and in the lexicon when the lexicon holds its words with its type; a prediction is classed the same way by its words and predicted type, and the precision of each part is the share of its unseen predictions that are correct. On MuLMS's decision seeds the names add 8,336 predictions of names the lexicon holds, 39% of them right, and 12,206 of other names, 22% right (419 more unseen-name predictions per run, 29% right), so two thirds of the wrong predictions they add are names the lexicon does not hold, and 45% of the added correct ones too; precision falls on both parts, 74.7% to 69.0% on the held names and 46.4% to 44.0% on the others. The mixed-corpus system's own lexicon holds 18.3% of the unseen test names, not the topic lexicon's 22.2% by which its rows are split.

The reading rule. Fixed before the run (31 October 16:13, `2c60f33`): "Name knowledge: the names' recall gain over MR6 is larger on the unseen names the lexicon holds than on those it does not, the paired interval of the difference above 0. A generic shift: the two gains are alike, the interval covering 0 with both gains positive. Mixed: the interval above 0 but the gain on the names not held also above 0." The rule does not say whether a gain is read on its point estimate or its interval. On MuLMS the difference's interval is above 0 (+6.17 [+2.44, +9.90]) and the gain on the names not held is +1.12 [−0.78, +3.02]: mixed on its point estimate, name knowledge on its interval. On MaterioMiner both gains are positive at both granularities and the difference's interval covers 0 (fine-grained +4.33 [−5.33, +13.98], coarse-grained −1.54 [−3.60, +0.51]): a generic shift on point estimates, and none of the three outcomes on intervals, every gain interval covering 0. The outcome note of 31 October read MuLMS

as name knowledge and MaterioMiner as a generic shift, the first on intervals and the second on point estimates; read one way throughout, the rule gives mixed and a generic shift, or name knowledge and none.

Table S3. Recall on the unseen test names by lexicon membership (post hoc; lexicons rebuilt by the names method's own teachers; MuLMS on the decision seeds, MaterioMiner on 10 seeds; a test name is unseen by the main text's words-only rule (Methods, *Data*); the mixed-corpus rows are split by the topic corpus's lexicon, the one rebuilt). Levels: per system, the share of the unseen test names the lexicon holds (pooled over seeds and units), recall on those and on the others, the share of the unseen-name predictions that are lexicon names, and precision and F1 on each part (F1 from the pooled counts, 2 correct / (gold + predicted)). Paired gains: per comparison, the recall gain on the held names and on the others, the difference of the two gains, paired over the units with 95% CIs, the units where the gain on the held names is the larger, and the F1 gain on the held names, each unit's F1 from its counts on them pooled over the seeds.

Levels

setting	system	unseen test names in the lexicon (%)	recall, in	recall, out	unseen predictions in the lexicon (%)	precision, in	precision, out	F1, in	F1, out
MuLMS, 7 subfields	CRF head	22.2	71.8	29.7	30.7	74.3	47.6	73.1	36.6
MuLMS, 7 subfields	MR3	22.2	75.6	34.1	28.4	74.1	46.3	74.8	39.2
MuLMS, 7 subfields	MR6	22.2	75.0	35.0	27.5	74.7	46.4	74.8	39.9
MuLMS, 7 subfields	MR6 + names	22.2	82.4	36.8	29.0	69.0	44.0	75.1	40.1
MuLMS, 7 subfields	MR6 + names (mixed corpus)	22.2	81.6	36.7	28.8	70.7	45.0	75.7	40.4
MaterioMiner fine, 4 papers	CRF head	9.8	74.0	30.8	13.0	81.3	46.4	77.5	37.0
MaterioMiner fine, 4 papers	MR3	9.8	83.1	34.0	13.0	82.7	46.3	82.9	39.2
MaterioMiner fine, 4 papers	MR6	9.8	82.5	35.0	12.3	81.4	44.6	82.0	39.2
MaterioMiner fine, 4 papers	MR6 + names	9.8	86.6	36.1	13.6	81.2	48.8	83.8	41.5
MaterioMiner coarse, 4 papers	CRF head	18.2	79.6	45.4	17.7	82.2	45.5	80.9	45.4
MaterioMiner coarse, 4 papers	MR3	18.2	79.1	47.2	17.3	82.8	46.6	80.9	46.9
MaterioMiner coarse, 4 papers	MR6	18.2	80.5	46.9	17.8	81.5	46.1	81.0	46.5
MaterioMiner coarse, 4 papers	MR6 + names	18.2	80.7	48.3	17.5	82.4	47.0	81.6	47.7

Paired gains

Table S3 (continued).

setting	comparison	recall gain on the names in the lexicon [95% CI]	recall gain on the names not in it [95% CI]	in − out [95% CI]	units with in > out	F1 gain on the names in the lexicon [95% CI]
MuLMS, 7 subfields	MR6 + names − MR6	+7.29 [+3.27, +11.30]	+1.12 [−0.78, +3.02]	+6.17 [+2.44, +9.90]	7/7	+1.13 [−1.85, +4.11]
MuLMS, 7 subfields	MR6 + names − CRF head	+10.67 [+5.14, +16.20]	+6.34 [+2.60, +10.07]	+4.33 [+1.18, +7.48]	6/7	+3.29 [−1.27, +7.85]
MuLMS, 7 subfields	MR6 − CRF head	+3.39 [+0.92, +5.85]	+5.22 [+1.92, +8.52]	−1.84 [−5.27, +1.60]	3/7	+2.16 [+0.15, +4.16]
MuLMS, 7 subfields	MR3 − CRF head	+3.56 [+1.11, +6.01]	+4.22 [+0.87, +7.57]	−0.66 [−2.78, +1.45]	3/7	+2.02 [−0.02, +4.06]
MuLMS, 7 subfields	MR6 + names (mixed corpus) − MR6	+6.16 [+1.32, +11.01]	+1.31 [−1.07, +3.68]	+4.86 [+1.50, +8.21]	7/7	+1.18 [−1.63, +4.00]
MaterioMiner fine, 4 papers	MR6 + names − MR6	+4.62 [−5.27, +14.50]	+0.29 [−4.23, +4.81]	+4.33 [−5.33, +13.98]	3/4	+1.96 [−6.54, +10.46]
MaterioMiner fine, 4 papers	MR6 + names − CRF head	+12.77 [−12.61, +38.15]	+6.00 [+2.84, +9.16]	+6.77 [−18.59, +32.13]	2/4	+6.57 [−18.79, +31.94]
MaterioMiner fine, 4 papers	MR6 − CRF head	+8.15 [−7.65, +23.95]	+5.71 [+1.05, +10.37]	+2.44 [−15.96, +20.85]	2/4	+4.61 [−12.46, +21.68]

setting	comparison	recall gain on the names in the lexicon [95% CI]	recall gain on the names not in it [95% CI]	in − out [95% CI]	units with in > out	F1 gain on the names in the lexicon [95% CI]
MaterioMiner fine, 4 papers	MR3 − CRF head	+8.77 [−5.95, +23.49]	+4.17 [−0.17, +8.50]	+4.60 [−13.82, +23.03]	3/4	+5.73 [−9.03, +20.50]
MaterioMiner coarse, 4 papers	MR6 + names − MR6	+0.36 [−1.11, +1.84]	+1.91 [−1.46, +5.28]	−1.54 [−3.60, +0.51]	0/4	+1.17 [−2.23, +4.58]
MaterioMiner coarse, 4 papers	MR6 + names − CRF head	+1.41 [−0.38, +3.19]	+3.62 [+1.18, +6.07]	−2.22 [−5.06, +0.62]	0/4	+1.47 [−1.36, +4.30]
MaterioMiner coarse, 4 papers	MR6 − CRF head	+1.04 [−1.72, +3.80]	+1.72 [−0.37, +3.80]	−0.67 [−1.53, +0.19]	0/4	+0.30 [−2.84, +3.43]
MaterioMiner coarse, 4 papers	MR3 − CRF head	−0.12 [−1.58, +1.34]	+2.32 [+0.88, +3.75]	−2.44 [−2.63, −2.24]	0/4	+0.62 [−1.51, +2.75]

The recall-matched control of Supplementary Note 10, registered after this analysis, tests one alternative to its reading, a uniform shift toward tagging more spans. The companion study's earlier recall calibration (Table S1, week 7 part 2a), a bias on the CRF head's emissions chosen on validation folds and not matched to the names, gave −0.59 [−1.75, +0.57] fine-grained and +0.17 [−0.14, +0.48] coarse-grained on MaterioMiner and −0.21 [−0.58, +0.17] on SOFC-Exp, paired by seed, the entities it added being right 19–22% of the time, against the names' net rate of 30% above.

Supplementary Note 6: per entity type

Table S4 gives, per type, the F1 of the CRF head, MR6 and MR6 + names on MuLMS's subfields, averaged over the (seed, unit) pairs whose test set has the type (the MaterioMiner tables, both granularities, are in the repository); these are post hoc and not decided on (Results, *Whose gain is it*). On MuLMS, MR6 + names gains over the CRF head on the 12 types with a non-zero score (RANGE, 3 entities per unit, scores 0 under every system), and MR6 holds more than half of that gain on 9 of them (not on MEASUREMENT, TECHNIQUE and DEV, where the names' own increment is +1.7 to +2.8). On MaterioMiner, coarse-grained, MR6 + names gains over the CRF head on 16 of 27 types and MR6 holds more than half of the gain on 13 of those; the names also lose against MR6 on several types, two where MR6 had gained (MEP −8.0, STD −7.0) and one where it had not (MAT −5.8), and most on types with a handful of test entities (PRP −10.2, CPR −9.2).

Table S4. Per entity type on MuLMS's 7 held-out subfields (decision seeds): the test support per unit, the number of units with the type, micro F1 of the CRF head, MR6 and MR6 + names, and the names method's differences; types ordered by test support.

type	test support per unit	units	CRF head	MR6	MR6 + names	names − CRF head	names − MR6
MAT	1806.7	7	53.92	56.94	57.50	+3.58	+0.56
VALUE	661.1	7	72.74	74.53	75.47	+2.72	+0.94
PROPERTY	554.3	7	28.97	32.36	33.60	+4.62	+1.24
FORM	454.3	7	24.97	31.83	32.51	+7.53	+0.68
MEASUREMENT	300.9	7	40.11	41.40	43.11	+2.99	+1.71
CITE	243.4	7	82.05	85.15	84.56	+2.51	−0.59
SAMPLE	198.7	7	3.61	5.66	4.33	+0.73	−1.33
TECHNIQUE	131.4	7	17.34	19.73	22.49	+5.15	+2.76
DEV	108.1	7	13.87	16.18	18.69	+4.83	+2.51
INSTRUMENT	52.9	7	17.69	22.11	20.21	+2.52	−1.90
NUM	37.1	7	5.57	8.11	8.71	+3.14	+0.60
UNIT	13.0	7	12.78	21.48	21.00	+8.21	−0.48
RANGE	3.0	6	0.00	0.00	0.00	+0.00	+0.00

Supplementary Note 7: the earlier protocols, the screen and other analyses

MR6 + names was reached through the protocols of weeks 10 and 11 and the screen of week 12, on MuLMS's held-out subfields and their MaterioMiner analogues, each committed before its runs and failing or passing as recorded:

- *Mention replacement alone* (week 10, 17 October 20:21): +2.10 [−0.31, +4.51] against the CRF head on MuLMS at 10 seeds, no; its votes against the CRF head's votes, +2.71 [+0.22, +5.20], pass. On MaterioMiner it had passed at both granularities (+2.77 and +1.28; Table S1, week 7 part 2b).
- *Names from the other subfields' unlabeled sentences* (week 11, 18 October 01:13): the three teachers tag the labeled subfields' sentences outside the pool, with a sentence-level filter (a sentence is silver when all its tags agree at 0.9), whose names hold 4.4% of a held-out subfield's unseen test names (1–8% per subfield): +2.04 [−0.85, +4.92] against the CRF head and −0.06 [−1.09, +0.96] against MR3; on MaterioMiner, with the fatigue corpus and the same filter, +0.23 [−0.69, +1.14] and +0.61 [−0.96, +2.18] against MR3 (2–9% of a paper's unseen names admitted). Four decisions, none passing; the route's reading at the time, that the filter and not the corpus was the bottleneck, is not borne out by the main text's ablations (Results, *The names' source and the lexicon's settings*).
- *Seven variants on the topic corpora* (week 12, 18 October 11:00, a 3-seed screen on the 7 subfields; Table S5, with the plan's labels, under which V5 was not assigned). The selection rule took the largest mean gain over MR3, which favors the noisiest variant; the screen decides nothing, and two of its losers deserve a 10-seed test that was not run: self-training (the student trains on the sentences the teachers labeled, not only on their names) on the corpus's silver sentences (V6) was the one variant whose interval against the CRF head cleared 0, by raising precision on unseen names (58.5 at 40.4% recall), and the lexicon as a gazetteer match feature of the CRF head, the classical use of a harvested lexicon, scored below MR3 on 3 seeds (V8's interval below 0, V7's covering it).

Table S5. The week-12 screen: seven variants on MuLMS's 7 held-out subfields at 3 seeds, micro F1 and the differences from MR3 and the CRF head (no decision).

Variant (3 seeds, 7 subfields; no decision)	Micro F1	Δ vs MR3 [95% CI]	Δ vs CRF head [95% CI]
MR6 + names, mention filter (V4, selected)	52.09	+1.45 [−1.35, +4.25]	+3.16 [−1.96, +8.28]
MR3 + names, mention filter (V2)	51.60	+0.97 [−0.91, +2.84]	+2.67 [−1.25, +6.59]
MR3 + names, sentence filter (V1)	51.44	+0.80 [−0.91, +2.51]	+2.50 [−2.09, +7.09]
self-training on the corpus's silver sentences (V6)	50.80	+0.16 [−2.31, +2.63]	+1.87 [+0.20, +3.53]
MR6, no names (V3)	50.44	−0.20 [−3.20, +2.80]	+1.50 [−1.65, +4.66]
lexicon as a gazetteer feature, mention filter (V7)	48.91	−1.73 [−4.13, +0.67]	−0.03 [−2.02, +1.97]
lexicon as a gazetteer feature, sentence filter (V8)	48.79	−1.85 [−3.59, −0.12]	−0.15 [−2.15, +1.86]

- *The lexicon as a dictionary* (a CPU probe planned on 19 October before it ran; no decision): applied to the CRF head's predictions as a rule, every verbatim match of a name tagged as its type where the head had tagged O, it helps where the head's recall trails its precision by 15 points or more (sofc +5.9, pemfc +4.0, semiconductors +2.3) and hurts elsewhere (steel −10.0, graphene −5.1), −1.0 on average over the seven subfields; its lexicons, from teachers trained on CPUs, differ from the runs' (25% of the unseen names on average against 23%), so it speaks to the dictionary rule only. Averaged over the 10 seeds it raised recall over all test entities in every subfield, by 1.4 (poly) to 11.2 points (sofc), and lowered precision by 2.8 (pemfc) to 24.2 (steel) (`results/silver_lexicons/`); recall on the held names was not scored. Supplementary Note 13 gives it per subfield with the sentence filter's lexicon (+1.13 on average) and the gold unseen names as a ceiling (+2.68 filling the head's O tags, +11.18 overriding them).
- *A second rule, label propagation within a document* (below): a term the model tags in most of its occurrences in a document is tagged in the rest.

Label propagation within a document. A post-processing rule on the saved predictions: a surface form that the model tags in most of its occurrences in a document, with one most frequent type, is tagged in its untagged occurrences too. On the CRF head's 10-seed predictions it gains +0.82 [+0.18, +1.46] on MuLMS's held-out subfields (pre-registered, every subfield gaining; 1,213 entities added per seed) and +0.12 [+0.01, +0.24] on coarse-grained MaterioMiner (pre-registered in an earlier week; fine-grained +0.30 [−0.35, +0.95], no), and loses −0.44 [−0.72, −0.16] on SOFC-Exp's article folds (pre-specified), where the annotators leave 14% of a tagged term's occurrences untagged on the official test set. It is not shown to add to mention replacement (+0.57 [−0.12, +1.25] on MuLMS). Two of its ten pre-registered decisions pass (six of 16 October, in the companion study's week 7; four of 20 October, in week 13, two on MuLMS registered at 09:14 and two on MaterioMiner at 09:25, after the MuLMS results, computed 22 minutes later from saved predictions), where about 0.25 would by chance.

The count. The twenty decisions counted in the main text (Results, *What else was tried, and the count*) are the twelve of Table 2; the six of the protocols of 17 October (20:21) and 18 October (01:13), on mention replacement alone on MuLMS (its single-model decision failed and its votes passed against the CRF head's votes) and on names from the other subfields' unlabeled sentences (four, none passing); and the two of 20 October (09:14) on mention replacement without names on SOFC-Exp's folds (MR3, as in the main text's Results, *SOFC-Exp: no gain shown*; and multi-task training with mention replacement, −0.11 [−0.69, +0.46]; both failing). Of the twenty, five passed: D3, D4 and D5 against the CRF head, D11 on recall (as registered, only with three subfields that no bias matched; +6.05 [−3.33, +15.44] on the four matched), and mention replacement's votes against the CRF head's votes. Of the twelve of Table 2, the seven against a mention-replacement control or on the names' source are D2, D7, D8, D9, D10, D11 and D12; only D11 passes, as registered, with that qualification. The label-propagation rule's decisions are not counted, the rule being a post-processing rule on saved predictions rather than a use of names; four of them fall in the window (20 October, above), and with them the count is twenty-four, six passing (13.1a, the rule's MuLMS decision of 09:14, among them), about 0.6 expected by chance. The companion study's decisions, all earlier, are not counted either.

Training size on SOFC-Exp. A decision of the companion study (17 October 07:28; Table S1, week 9) trained on 10 of the official split's 26 training articles (5 draws of 10 articles, 156–248 sentences and 820–1,276 entities each, about as many entities as MaterioMiner's training sets hold; 10 seeds each) and scored on its test articles: MR3 gained +0.50 [−0.47, +1.46] over the CRF head and did not pass. MR6 + names was not run at that size.

Supplementary Note 8: the follow-up on the strongest system in detail

Design. Table S6 gives the follow-up's comparisons. The control, multi-task training with six-copy mention replacement, and the names system's two-head filter are described in the main text's Methods (*Systems*); the three multi-task teachers are the stored multi-task runs of seeds 0, 100 and 200, whose scores the jobs reproduced. The protocol expected detectable differences of about 3.4 to 3.8 points fine-grained and 1.4 to 2.4 coarse-grained.

Table S6. The follow-up: the names on the strongest MaterioMiner system. Micro F1 over the four held-out papers (10 seeds; both granularities from one model) with its 95% CI; the names against the matched control (decisions D9 and D10), and both against multi-task training with three-copy mention replacement, the multi-task model and the CRF head (pre-specified). A bound of −0.00 lies just below 0.

	System	micro F1 [95% CI]	Δ vs the control	Δ vs multi-task + MR3	Δ vs multi-task	Δ vs CRF head
fine	multi-task + MR6 + names	53.95 [47.52, 60.38]	+0.96 [−0.01, +1.92] (D9, no)	+0.76 [−0.15, +1.68]	+3.77 [+1.34, +6.21]	+7.49 [+5.96, +9.02]
fine	multi-task + MR6 (the control)	52.99 [46.93, 59.06]		−0.19 [−2.05, +1.67]	+2.82 [−0.00, +5.64]	+6.54 [+4.14, +8.94]
coarse	multi-task + MR6 + names	61.25 [53.71, 68.80]	+0.85 [−0.72, +2.42] (D10, no)	+0.91 [−0.14, +1.97]	+3.21 [+0.46, +5.96]	+2.65 [−0.09, +5.38]
coarse	multi-task + MR6 (the control)	60.40 [53.66, 67.15]		+0.06 [−0.74, +0.86]	+2.36 [+1.15, +3.56]	+1.80 [+0.19, +3.40]

Lexicons. The two-head filter admitted lexicons of 16,455 to 42,066 entries per paper (3,746 to 6,587 distinct fine-typed names). By the runs' stored counts, which call a test name unseen when no training sentence annotates its words with its type, they hold 7.5% of the unseen test names fine-grained and 12.1% coarse-grained, pooled over the papers, against 9.6% and 18.3% for the single-task lexicons of D3 and D4, lower in 7 of the 8 paper–granularity cells (per paper 6.5 to 16.3% of the unseen fine-grained test names and 6.7 to 24.6% of the coarse-grained, against 8.4 to 18.6% and 15.5 to 23.3%); Table 5's 9.8% and 18.2% count by the main text's words-only rule (Methods, *Data*).

Unseen names. On the unseen test names the names' changes over the multi-task control are +0.58 [−2.93, +4.09] recall and +0.82 [−3.39, +5.03] precision fine-grained, and +1.45 [−0.69, +3.58] and +0.97 [−1.09, +3.03] coarse-grained (pre-specified; paired over the four papers). Post hoc, F1 changes by +2.43 [+0.40, +4.47] on the seen and +0.43 [−1.57, +2.44] on the unseen test names fine-grained, and by +0.02 [−1.30, +1.34] and +1.20 [−0.82, +3.22] coarse-grained; no pattern is read from these.

Cost. A names job took 82.5 to 122.7 minutes on a T4 (by the results files' run times; three teachers, their tagging of 40,000 sentences with two heads, and ten students): 10.0 minutes per student with the harvest shared over the job's ten students and about 38 charged to one, against 6.9 for a control run and about 2.4 for the two CRF-head runs that give both granularities (Supplementary Note 2).

Supplementary Note 9: detectable differences

Table S7 gives the smallest true mean difference each decision and main comparison detects with 80% power at the two-sided 5% level, from the half-width of its observed interval and its degrees of freedom (`ner_robustness.py`, `detectable`).

Table S7. The difference each comparison could detect at 80% power and the two-sided 5% level, from its stored interval's half-width and degrees of freedom. These rest on the observed spread and are uncertain themselves: the 95% interval of a standard deviation runs from 0.57 to 3.7 times its estimate with 3 degrees of freedom, 0.64 to 2.2 times with 6 and 0.70 to 1.8 times with 10.

Comparison	CI half-width	df	detectable at 80% power
D5: MR6 + names − CRF head, pooled, 10-seed basis	2.63	10	3.67
D5′: the same on the decision seeds	2.54	10	3.55
D7: MR6 + names − MR6, pooled	1.46	10	2.04
MR6 + names − MR3, pooled (post hoc, decision seeds)	1.63	10	2.28
MR6 − MR3, pooled (post hoc, decision seeds)	0.58	10	0.82
MR3 − CRF head, pooled (post hoc, decision seeds)	1.46	10	2.03
D1: MR6 + names − CRF head, MuLMS	4.32	6	5.92
D2: MR6 + names − MR3, MuLMS	2.55	6	3.50
MR6 + names − MR6, MuLMS (D7's per-corpus condition)	2.02	6	2.77
D8: own-topic − mixed corpus, MuLMS	1.83	6	2.51
D3: MR6 + names − CRF head, MaterioMiner fine	2.88	3	3.77
D4: MR6 + names − CRF head, MaterioMiner coarse	1.10	3	1.44
MR6 + names − MR6, MaterioMiner fine (D7's per-corpus condition)	3.80	3	4.97
the same, coarse (pre-specified)	2.30	3	3.01
D6: MR6 + names − CRF head, SOFC-Exp, 5 folds	1.17	4	1.57
D9: multi-task + MR6 + names − multi-task + MR6, MaterioMiner fine (follow-up)	0.96	3	1.26
D10: the same, coarse (follow-up)	1.57	3	2.05
D11: recall on the held names, MR6 + names − the recall-matched control, MuLMS (follow-up)	7.93	6	10.87
D12: micro F1, the same (follow-up)	4.05	6	5.55
D11 without the flagged subfields (pre-specified)	9.38	3	12.27
D12 without the flagged subfields (pre-specified)	2.75	3	3.60
D11, the flagged subfields at the nearest recall (post hoc)	5.28	6	7.23
D12, the same (post hoc)	2.04	6	2.79
multi-task + MR6 + names − multi-task + MR3, MaterioMiner fine (pre-specified)	0.91	3	1.20
the same, coarse (pre-specified)	1.05	3	1.38
ablation, 10,000 sentences − MR6 + names (3 seeds)	1.63	6	2.23
ablation, 20,000 sentences − MR6 + names (3 seeds)	2.29	6	3.14
ablation, threshold 0.7 − MR6 + names (3 seeds)	1.47	6	2.01
ablation, 2 of 3 teachers − MR6 + names (3 seeds)	1.70	6	2.32
ablation, names in half of the draws − MR6 + names (3 seeds)	1.71	6	2.35

Supplementary Note 10: the recall-matched control in detail

Design. Registered on 1 November at 10:51 (`e02c3be`), after an internal review of the draft asked whether the names' recall gain on the unseen names their lexicon holds (Results, *What the names do*) is name knowledge or a generic shift toward tagging more spans. The control (MR6's stored decision jobs run again and decoded under the bias), the matched bias D*, the flagged subfields and the two decisions are defined in the Methods (*Systems*, *The recall-matched control*); the decisions apply the main text's decision rule (Methods, *Evaluation protocol*). The protocol also fixed what the abstract could say: "18.1 passes: … paper A's abstract may state the recall finding with this decision. 18.1 fails: that is not shown, and the finding stays out of the abstract." D11 (18.1) passed as registered, and the abstract states the recall finding with it and its qualification, the three subfields the bias could not match. The protocol expected D12 near +0.7 to +1.3 (the names' F1 over MR6 at D = 0 on these units is +0.51

[−1.51, +2.53], and on MaterioMiner a bias from 0 to 1 had cost the CRF head about 0.2 F1 for 2 points of recall), a pass unlikely; it predicted no number for D11.

Checks. The session's repeat of an earlier CRF-head run was identical; at D = 0 all 49 control runs equal their stored MR6 runs in F1, precision, recall, best epoch, training history and test predictions; the lexicons are the rebuilt ones of the main text's lexicon split (Results, *What the names do*), whose checks pass.

The matching. Table S8 gives the control's recall at every bias. D* is −1.08 for graphene, where the names' recall is below MR6's, and 1.35 to 2.47 for poly, steel and semi. In electrolysis, pemfc and sofc MR6's recall rises to a peak at D = 2.5 or 3 (59.66, 51.54 and 49.93), 0.47, 0.20 and 0.43 points short of the names' (60.13, 51.74 and 50.36), and falls beyond it: from D = 2.5 to 5 in electrolysis the bias adds about 9% more predicted entities while the correct ones fall by 4%, spans growing or merging until they no longer match exactly. The protocol's rule, which presumed recall to rise with D, matched these three at D = 5, where MR6's recall is 57.22, 50.03 and 47.57 and its seed-mean F1 5.3 to 7.0 points below its own at D = 0 (5.4 to 7.0 in Table S8's pooled F1).

Further comparisons (pre-specified, at D*, paired over the seven subfields). Precision on all test entities +5.78 [−1.22, +12.79]; on the unseen test names, recall +1.12 [+0.09, +2.15], precision +6.17 [−1.27, +13.61] and F1 +3.54 [−0.58, +7.67]; recall on the unseen names the lexicon does not hold −1.73 [−3.02, −0.44], positive in one subfield; the difference of the two parts' differences +12.98 [+4.18, +21.77]; the bias's own F1 change at D*, −2.75 [−5.77, +0.27]. Without the flagged subfields D11 is +6.05 [−3.33, +15.44], D12 −0.03 [−2.78, +2.72] and precision +0.56 [−6.74, +7.85].

Cost. The control's seven jobs took 31.9 to 39.4 minutes each on a T4, a training run and twelve decodes per subfield and seed, in one session of 151 minutes.

Post hoc, after the result. Four analyses were added after the result, the first on the day of the result and the others after an internal review of the draft asked for them; all are labeled post hoc in `ner_recall_matched.py` and its results file. (1) With the flagged subfields matched at the grid value of the nearest recall (2.5, 3 and 2.5; `match_nearest`), D11 is +8.63 [+3.35, +13.90], positive in every subfield, D12 +1.37 [−0.66, +3.41], precision +3.04 [−1.32, +7.41], recall on the unseen names the lexicon does not hold −2.07 [−3.68, −0.46] and on all unseen names +0.25 [−0.08, +0.58]. (2) The rates of the control at every bias on the unseen names in and out of the lexicon (Table S8; Figure 4 shows those in the lexicon): MR6's recall on the held names is highest at D ≤ 0.5 in five of the seven subfields and falls as D grows, while its recall on the other unseen names rises; at whichever bias gives MR6 its highest recall on the held names, the names lead by +6.52 [+2.15, +10.89], in every subfield, and on the other unseen names a bias overtakes them, −4.16 [−5.82, −2.50] (none of seven above 0). (3) The decomposition of D11: per subfield, the names' gain over MR6 at D = 0 on the held names (+7.29 [+3.27, +11.30], Table 5) plus MR6's own loss on them from D = 0 to D* (+3.96 [−1.06, +8.98]); on the four matched subfields the two parts are +5.92 [−1.10, +12.94] and +0.14 [−4.89, +5.16], so there matching the recall leaves the gain where it was at D = 0. (4) The split by type (Table S9). Values and materials names are 83 to 99% of a subfield's distinct lexicon names and 83 to 99.6% of its entries (Table S10), and they are 60 to 99.6% of the unseen test names a lexicon holds and 24 to 59% of the others (a minority in five of the seven subfields, 46% pooled), so a shift toward those two types could favor the held names without any name being learned. At D = 0, where no matching is involved, the materials names the lexicon holds gain +7.98 [+2.15, +13.82] over MR6 (7 of 7 subfields) and the other materials names +0.89 [−3.48, +5.27]; values held +1.74 [−1.16, +4.65], the other values −0.23 [−3.00, +2.53]; names of the other types held +9.80 [+3.34, +16.25], from cells of 6 to 577 names, and not held +1.38 [−0.82, +3.58]. Of the held names recalled beyond MR6, 68% are materials names, 27% of the other types and 5% values (per seed, summed over the subfields). At D* the same split reads +12.02 [+3.70, +20.34] for the held materials names (+9.87 [+3.57, +16.16] with the flagged subfields at the nearest recall), −1.19 [−5.59, +3.21] for the other materials names, +9.43 [+0.02, +18.83] and +3.41 [−1.97, +8.80] for values, and −3.91 [−7.32, −0.51] for the other types not held, the matched bias adding its own loss on the held names. The pattern fits gains specific to the names the lexicon holds, chiefly materials names, with little on values; the shift away from the types the lexicon barely holds appears only against the bias-shifted control. The exact two-sided sign-flip P over the seven

subfields is 0.016 for D11, the smallest seven units allow, and 0.14 for D12; at 80% power D11 detects 10.87 points (12.27 without the flagged subfields, 7.23 post hoc) and D12 5.55 (3.60 and 2.79; Table S7); the two-sided t-test P values are 0.013 for D11 and 0.096 for D12.

Table S8. The recall-matched control's rates at every bias D, pooled over the decision seeds, beside MR6 + names' (MuLMS's seven held-out subfields; `docs/tables/table12_week18.md`): recall, precision and F1 over all test entities (pre-specified), then recall on the unseen test names the lexicon holds and on the other unseen names (post hoc).

Subfield	rate	names	D = −2	D = −1	D = −0.5	D = 0	D = 0.5	D = 1	D = 1.5	D = 2	D = 2.5	D = 3	D = 4	D = 5
electrolysis	recall	60.13	54.35	56.44	57.37	58.17	58.78	59.20	59.46	59.61	59.66	59.59	58.81	57.22
electrolysis	precision	56.40	61.91	59.89	59.04	58.08	57.18	56.14	55.04	53.88	52.66	51.44	48.86	46.13
electrolysis	F1	58.21	57.88	58.11	58.19	58.12	57.97	57.63	57.16	56.60	55.94	55.22	53.38	51.08
graphene	recall	47.45	46.25	47.56	48.22	48.79	49.39	49.90	50.41	50.66	50.93	51.19	51.42	51.64
graphene	precision	54.51	59.56	57.39	56.55	55.53	54.62	53.59	52.76	51.86	50.90	50.15	48.41	46.54
graphene	F1	50.74	52.07	52.01	52.06	51.94	51.87	51.68	51.56	51.26	50.92	50.66	49.87	48.96
pemfc	recall	51.74	44.18	47.23	48.31	49.23	49.93	50.50	50.93	51.24	51.41	51.54	51.15	50.03
pemfc	precision	54.95	59.94	58.04	56.80	55.46	54.24	53.08	51.92	50.77	49.63	48.44	46.14	43.91
pemfc	F1	53.30	50.87	52.08	52.21	52.16	51.99	51.76	51.42	51.00	50.50	49.94	48.51	46.77
poly	recall	53.52	45.69	48.39	49.53	50.61	51.83	52.85	53.80	54.39	54.93	55.36	55.19	53.98
poly	precision	45.22	58.40	55.36	53.65	52.02	50.57	48.98	47.29	45.66	44.03	42.49	39.29	36.46
poly	F1	49.02	51.27	51.64	51.51	51.31	51.19	50.84	50.34	49.64	48.88	48.08	45.90	43.52
semi	recall	31.78	21.19	24.14	25.53	26.82	27.99	29.09	30.16	31.01	31.84	32.37	32.19	31.27
semi	precision	50.85	62.73	58.48	56.21	54.23	52.03	49.89	47.86	45.77	43.83	41.90	37.50	33.73
semi	F1	39.11	31.68	34.17	35.11	35.89	36.40	36.75	37.00	36.97	36.88	36.52	34.64	32.45
sofc	recall	50.36	40.22	43.79	45.26	46.55	47.71	48.59	49.29	49.85	49.93	49.87	49.54	47.57
sofc	precision	60.82	63.37	61.02	59.77	58.50	57.28	56.12	54.89	53.56	52.00	50.42	47.45	43.89
sofc	F1	55.10	49.21	50.99	51.51	51.84	52.06	52.08	51.94	51.64	50.95	50.14	48.47	45.65
steel	recall	56.37	49.78	51.99	52.92	53.83	54.57	55.30	55.87	56.46	57.00	57.46	58.08	58.00
steel	precision	57.10	66.83	64.29	62.98	61.77	60.29	58.88	57.35	55.95	54.52	53.08	50.24	47.29
steel	F1	56.73	57.06	57.49	57.51	57.53	57.29	57.04	56.60	56.21	55.73	55.18	53.88	52.10
electrolysis	recall, held names (post hoc)	90.02	82.34	83.17	83.54	83.62	83.44	83.10	82.62	81.91	81.26	80.60	78.65	76.73
electrolysis	recall, other unseen names (post hoc)	38.23	33.14	36.19	37.61	38.84	39.97	40.96	41.66	42.25	42.80	43.19	43.12	42.01
graphene	recall, held names (post hoc)	84.91	84.00	84.75	84.69	84.43	84.26	84.00	83.94	83.46	82.98	82.76	81.90	81.15
graphene	recall, other unseen names (post hoc)	30.09	27.85	29.73	30.81	31.66	32.60	33.53	34.42	35.00	35.65	36.24	36.88	37.54
pemfc	recall, held names (post hoc)	84.92	75.86	77.63	77.95	78.09	77.74	77.19	76.39	75.58	74.55	73.74	71.89	69.33
pemfc	recall, other unseen names (post hoc)	38.19	31.47	34.86	36.18	37.26	38.27	39.16	39.92	40.51	41.03	41.43	41.42	40.71
poly	recall, held names (post hoc)	71.22	61.57	64.51	65.47	66.58	67.65	68.65	69.56	69.83	70.02	70.07	69.41	67.52
poly	recall, other unseen names (post hoc)	38.98	32.08	34.67	35.91	36.95	38.31	39.33	40.35	41.04	41.77	42.40	42.43	41.46
semi	recall, held names (post hoc)	81.36	67.15	71.58	73.32	73.58	73.35	73.12	72.16	70.50	68.85	66.88	62.03	57.37
semi	recall, other unseen names (post hoc)	26.87	16.35	19.22	20.63	22.02	23.29	24.52	25.76	26.78	27.75	28.49	28.59	27.93
sofc	recall, held names (post hoc)	85.32	67.13	69.86	70.82	71.23	71.38	70.97	70.49	70.10	69.09	67.59	64.89	59.72
sofc	recall, other unseen names (post hoc)	42.13	33.55	37.43	39.06	40.62	42.03	43.25	44.22	45.03	45.31	45.57	45.75	44.42
steel	recall, held names (post hoc)	84.39	70.65	72.69	73.29	73.61	73.96	74.25	74.45	74.77	74.88	75.01	74.69	73.98
steel	recall, other unseen names (post hoc)	43.70	39.06	41.14	42.04	43.02	43.85	44.72	45.32	45.97	46.61	47.18	48.19	48.40

Table S9. Recall on the unseen test names by type group and lexicon membership (post hoc): the names minus the control at D*, with the flagged subfields at the nearest grid value instead, and minus MR6 at D = 0, paired over the seven subfields with the subfields above 0; then the unseen test names per subfield in each cell, and the names MR6 + names recalls beyond MR6 at D = 0 in each cell, per seed summed over the subfields (each subfield's gain times its names), with, for the held names, whose cells all add names, the cell's share of their total.

Type	unseen names	names − control at D*	the flagged at the nearest recall	names − MR6 at D = 0	test names per subfield	names recalled beyond MR6 at D = 0, per seed
MAT	held	+12.02 [+3.70, +20.34], 7/7	+9.87 [+3.57, +16.16], 7/7	+7.98 [+2.15, +13.82], 7/7	184–1022	+316 (68%)
MAT	not held	−1.19 [−5.59, +3.21], 4/7	−1.58 [−5.61, +2.46], 2/7	+0.89 [−3.48, +5.27], 5/7	129–2223	+175
VALUE	held	+9.43 [+0.02, +18.83], 6/7	+5.61 [+0.27, +10.94], 6/7	+1.74 [−1.16, +4.65], 4/7	56–313	+23 (5%)
VALUE	not held	+3.41 [−1.97, +8.80], 4/7	+1.14 [−2.30, +4.58], 3/7	−0.23 [−3.00, +2.53], 3/7	106–681	−17
other	held	+2.87 [−4.82, +10.55], 5/7	+0.94 [−5.84, +7.72], 4/7	+9.80 [+3.34, +16.25], 6/7	6–577	+126 (27%)
other	not held	−3.91 [−7.32, −0.51], 1/7	−3.58 [−6.49, −0.67], 1/7	+1.38 [−0.82, +3.58], 6/7	754–2516	+222

Table S10. The rebuilt lexicons of MuLMS's subfields by type: distinct names and, in parentheses, entries (occurrences in the corpus, which set how often a name is drawn; counts only, the names are not released).

Subfield	VALUE	MAT	PROPERTY	MEASUREMENT	INSTRUMENT	TECHNIQUE	FORM	DEV	SAMPLE	CITE
electrolysis	7,168 (13,635)	2,392 (29,748)	506 (1,507)	121 (676)		70 (102)	17 (405)	15 (63)		1 (1)
graphene	6,089 (11,328)	1,553 (16,412)	28 (46)	72 (566)	60 (60)	2 (2)	7 (36)		2 (3)	
pemfc	4,928 (9,062)	357 (3,811)	10 (11)	3 (4)		2 (2)	3 (18)			18 (19)
poly	7,478 (15,546)	3,664 (26,517)	1,359 (3,098)	316 (3,528)	236 (239)	192 (386)	84 (1,438)	37 (80)	29 (37)	
semi	4,580 (7,429)	132 (666)	96 (116)	20 (115)			1 (1)			1 (1)
sofc	4,011 (8,359)	265 (5,167)	38 (41)	42 (153)	3 (3)		20 (678)			1 (1)
steel	5,797 (12,164)	1,492 (15,248)	439 (791)	224 (1,485)	23 (24)	35 (41)	48 (571)	3 (3)		5 (5)

Supplementary Note 11: checks, deviations, failures and timings of the names route

This Note gives, week by week (weeks 15 and 18 in Supplementary Notes 5 and 10), the checks the runs of the names route passed, where the runs departed from what the protocols planned, which sessions failed, and when each of these happened; commit times are the commits' (UTC), push and pull times the session notes', and a session's end or length its Kaggle manifest's, as in Supplementary Note 1.

Week 10. The protocol of 17 October 20:21 (`a08356f`) listed the analyses week 10 would report, and looser matching was not among them. The looser-matching script had been introduced on 16 October as an exploratory analysis. It was adapted to MuLMS's protocols on 17 October at 21:43 (`6979420`), after `w10a` had been pushed at 20:57, and at 21:55, before the results, a dry run on files from a tiny random model trained on MuLMS's real folds and splits went through the matching analysis. Week 10's result was recorded on 18 October at 00:48 (`c2ca3f6`). On the held-out subfields the CRF head scored 48.66 with span and type exact, 54.59 on the span alone and 69.06 under any overlap, against 65.72, 70.22 and 81.68 on the random pool splits, so the gap of 17.06 points fell to 12.62 under overlap. Of its predictions on the held-out subfields, 13.1% lay inside a longer gold entity, against 9.0% on the random splits.

Week 11. In `w11a` (MuLMS; pushed on 18 October at 01:50, done at 04:42 and pulled at 04:44), per subfield both lanes' (a lane is one of a session's two T4 GPUs, each running its own list of jobs) three teachers and silver counts were identical, and every teacher scored exactly as the stored week-10 CRF-head run of its seed. In `w11b` (MaterioMiner) each lane ran one granularity, and per paper and granularity the three teachers scored exactly as the stored CRF-head runs of their seeds. That a student trained without silver names equals the `crf_mr` run of its seed was shown by a test on a tiny model, committed with the week's preparation on 18 October at 01:12 (`50c07dc`), not by a GPU run. At 01:32, before the pushes, one `w11a` job (graphene) and one `w11b` job (coarse-grained, paper s11340) ran on the real data with a tiny random model on the CPU; in the `w11a` job the random model found no silver sentence and the teacher of seed 0 reproduced week 10's tiny-model CRF run on graphene (1.11), and the `w11b` job's 963 copies equalled those of week 7's mention-replacement run of seed 0 on that paper. On a tiny random model, every existing option that the week-11 change touched wrote byte-identical results files.

The protocol of 18 October 01:13 (`427e34c`) ran `w11b` after the weekly GPU quota's reset on 24 October, and `w11a` left about 1 GPU hour until then. The quota was raised to 45 hours a week on 19 October, and `w11b` was pushed that day at 21:06 to the Kaggle notebook `ner-gpu-jobs-d` (version 6), before the reset. It ran 3.4 hours (9 jobs), and its result, decisions 3 and 4 of week 11 failing, was recorded on 20 October at 00:44 (`04764ef`). The protocol named its own commit as every batch's source; `w11b` ran from bundle `77c37be`, the commit of week 12's first amendment, and its files carry the `silver_filter` field that the week-12 code added, which `w11a`'s files lack. The week-12 code of 18 October 11:00 (`40631ce`) had been checked to repeat the week-11 code's runs byte for byte apart from `runtime_sec`, its silver-name runs differing only by that field; no such check of `77c37be`'s own change to `ner_baseline.py` is recorded.

On 18 October at 04:52 (`fa64ed6`), after `w11a`'s result had been committed at 04:50, the plan made `w11b` the gate of the route: topic corpora for MuLMS only if decision 3 or 4 of week 11 passed, and otherwise the evaluation paper. Phase 0, measured on the CPU from 09:14 that day (`8b9dd59`), was to decide whether topic corpora were worth the GPU hours. The week-12 protocol, committed at 11:00 (`40631ce`), made phase 0's coverage the gate instead of `w11b`, which it said could not run before the reset, and kept `w11b` first with its decisions as fixed. When `w11b`'s failure was recorded, at 00:44 on 20 October, the week-12 protocol and its first two amendments were in place, and the week-12 screen was pushed at 02:35. Under the plan of 04:52 that failure would have led to the evaluation paper.

Week 12. Phase 0's first text, on 18 October at 09:14 (`8b9dd59`), tested only the SOFC corpus of the week-6 amendment, removed its overlap with MuLMS's sofc documents by a 30-character shingle (665 sentences, read then as some of those documents being among its articles), and put about 45 points of the coverage down to

generic materials names and about 35 points to the topic. At 11:00 (`40631ce`), with the seven topic corpora built, the generic part was 45–65 points and the own topic added 4–11 points on six subfields, while on graphene the own corpus was 3 points behind the electrolysis corpus. The 30-character shingle had matched technical terms in 270–319 of each topic's articles and dropped 62–224 articles per topic whole; it was widened to 50 characters, at which 15 sentences and no article of the SOFC pool shared a run with MuLMS, and at 60 none. No sentence of a topic corpus shared a 50-character run with a held-out subfield's documents; the SOFC and fatigue corpora, used as controls, shared 0 to 5 sentences with a subfield's documents, which the counts left out. MuLMS ships no article identifiers: 16 of its 50 documents were found on Europe PMC by title and excluded by PMCID, the rest coming from journals Europe PMC does not index, and every sentence sharing an 8-word n-gram or a 50-character run with a MuLMS sentence was dropped as well. For MuLMS, therefore, the exclusion "by identifier" of the Methods (*Systems*) covered 16 of its 50 articles.

Amendment 1 was committed on 19 October at 17:15 (`77c37be`), before any run, adding V4 (`crf_mrm_topic6`) and V6 (`crf_st3_topic`) to the screen. Amendment 2, V7 (`crf_gzm_topic`), was proposed at 20:54 (`48f37d3`) and recorded as not adopted and the user's call. It was adopted nine minutes later, at 21:03 (`8a18962`), before any run. Amendment 3, V8 (`crf_gzs_topic`), was committed on 20 October at 01:50 (`47e2310`) on the CPU probe's first six subfields, graphene pending, and kept the threshold at 0.9. The bundle rebuilt from `47e2310` was ready at 02:02, and the screen was pushed at 02:35 without waiting for graphene. Graphene's probe and the seven-subfield reading were committed at 07:01 (`dcbff64`). On graphene the sentence filter's 0.9 lexicon filled to −2.86, so that over the seven subfields it averaged +1.1 and reached −2.9, where amendment 3 had priced it on six at +1.8 on average and never below −0.3. The seven-subfield reading changed neither filter nor threshold.

The protocol required, before any number was used, identical smoke repeats, each job's teachers equal to week 10's `crf_m0`, `_m100` and `_m200` on their subfield, the bundle's corpora matching the reports' SHA-256 (checked again by the runner), and every run recording `silver_filter` and `names`. Every teacher of every job equalled the stored CRF-head run of its seed in `w12c` (version 4), in `w12e` and in `w12b:mrm6`, and on MaterioMiner every teacher of `w12d:mrm6` equalled the stored week-5 run of its seed on its paper. From the resume fix on, every run recorded its own corpus's SHA-256. At the reset of 24 October the bundle's `BUNDLE.json` named `3ac85a8`, the SOFC corpus's SHA-256 (1b6a00d2) and the seven topic corpora, and `ner_baseline.py`, `ner_consistency.py` and `kaggle/` were unchanged since, so the decision batches ran from it.

On a tiny model and the tests' toy corpora, the code of 18 October 11:00 (`40631ce`) repeated the week-11 code's runs byte for byte apart from `runtime_sec` for the CRF head, MR3, multi-task MR3, self-training and MuLMS's pool and held-out runs, and its silver-name runs differed only by the added `silver_filter` field, as recorded at 11:05 (`bd20cec`). On 19 October the XLA port's CPU path repeated the pre-port files byte for byte apart from `runtime_sec` on every tiny-model option. With amendment 2's feature, the paths it did not touch (`crf`, `mr`, `mt_mr`, `st3`, `mrs`, the MuLMS pool and unused-corpus runs) were byte-identical to `77c37be` on the tiny corpora. With the resume fix of 20 October (`3ac85a8`), the files of the other protocols were byte-identical to `77c37be` in 8 configurations, and a tiny end-to-end run showed two subfields extending one file, a repeat resuming and another corpus refused.

The screen was pushed on 20 October at 02:35 from bundle `47e2310`, `w12a` to `ner-gpu-jobs-d` (version 7) and `w12c` to `ner-gpu-jobs-c` (version 3). From 03:03 (`w12c`) and 03:11 (`w12a`) to 03:34 and 03:50 their later jobs failed on the resume check (the session notes recorded it from 03:56): a results file stored one corpus SHA-256, each held-out subfield had its own topic corpus, and the second subfield's job refused to resume a file "from another corpus". `w12c` completed in 59 minutes with 4 of its 14 topic jobs done and 10 failed in 0.01 minutes; `w12a` lost every names job but the first of each variant, while its `crf_mr6` jobs ran. `w12c` had also put each feature variant's jobs in both lanes, which would have had the lanes write the same files at once. The fix, committed at 04:09 (`3ac85a8`), made a run with `--subfields` record its own corpus, kept other subfields' runs in a file, resumed a subfield's run only from the same corpus and put one variant per lane. The bundle was rebuilt from `3ac85a8` at 04:20, and `w12c` was pushed again at 04:49 (version 4). The first `w12a` session ended at 04:53 (137.6 minutes, 2.3

quota hours) with the smoke repeat, electrolysis for V1, V2, V4 and V6 (29–38 minutes each) and the three `crf_mr6` jobs, and was pulled at 05:03. Of its files only the three `crf_mr6` files were collected, at 08:06; V1, V2, V4 and V6 ran again on all seven subfields, electrolysis included, as `w12e`. Electrolysis runs of V4, MR6 + names, on the screen's seeds had thus been pulled, though not collected or recorded, before the criterion was revised at 09:14. The record does not show whether their scores were read. Compared on 1 November with the pulled files, each of those twelve runs (V1, V2, V4 and V6 on electrolysis, three seeds each) is identical to its rerun in `w12e` and to the committed file in every score, the training history and the predictions; only the run time differs, and the first session's runs lack the per-run corpus hash that the resume fix (`3ac85a8`) added, the corpus being the same by that session's manifest. V4's electrolysis gain over the CRF head on those seeds is +2.06, below MR6's +2.97 (the electrolysis values of Table S11's rows 24 and 26, given per unit in Table S12).

`w12e` was pushed at 05:03 to `ner-gpu-jobs-d` (version 8) before the quota was checked against the job times: it needed about 8.1 hours and `w12c` 3.1, against 9.14 hours left in the week, and the notes expected Kaggle to stop a session when the quota ran out and keep none of its outputs. The user was asked to cancel version 8, and the plan, committed at 05:06 (`9e9d906`), split the names variants into `w12f` (V1 and V2), to run in its place, and `w12g` (V4 and V6), to run after the reset of 24 October with the decision batch. At 08:10 the quota left 2.9 hours, the notes recorded that canceling no longer changed the plan, and `w12e` was to run after the reset. The week-13 protocol, committed at 09:14, also put `w12e` after the reset, beside `w13a`. `w12e` was not canceled and `w12f` was never pushed: the next sessions on either Kaggle notebook (`ner-gpu-jobs-d`, `ner-gpu-jobs-c`) were those of 24 October, `ner-gpu-jobs-d`'s version 9 and `ner-gpu-jobs-c`'s version 5. Kaggle did not stop the session: the quota read 43.99 hours used at 10:10, 45.25 at 11:25 with the session running, and 46.23 of 45 at 13:23, when `w12e` was found complete; by its manifest it had ended at 12:25 (441 minutes, 28 jobs after the repeat). The 1.23-hour overrun was not carried into the next week. V4's screen result was therefore recorded on 20 October at 13:35 (`5f844bd`), before the amendments of 13:53 (`5f7f827`) and 18:42 (`de62446`); under the plans committed at 05:06, 08:11 and 09:14 it would have come after the reset of 24 October.

The reset of 24 October reached the session at 12:16, 12 hours late, because the container had restarted. `w12b:mrm6`, pushed at 12:22 (`ner-gpu-jobs-d`, version 9), failed without a trace at 13:44, after 82 minutes: status ERROR, an empty log, no manifest and no outputs, the API's failure message empty, about 1.4 GPU hours charged. It was pushed again unchanged at 13:49 (version 10), with its live log kept every 20 minutes, and completed at 18:09 (259.8 minutes). `w12d:mrm6`, pushed at 18:10 (version 11), still had an empty live log at 20:05 and was taken to have hung before its setup. Version 11 ended in ERROR at 20:11 with an empty log and no manifest; the quota then read 12.05 hours used, 1.8 fewer than at 20:04, so its hours were not charged. It was pushed again at 20:12 (version 12) and completed on 25 October at 00:48 (276.1 minutes; pulled at 00:56).

A session note headed 20:25 on 24 October, committed at 20:18 (`2c2409d`) and on GitHub at 20:18:40, before `w12d:mrm6`'s results, worked out what D5 needed. V4's 10-seed MuLMS units against the CRF head were +2.39, +1.75, +2.94, −2.76, +6.20, +12.16 and −0.40 (mean +3.18, standard deviation 4.84). With the four MaterioMiner papers spread as MR3's were there (standard deviation 0.8), the pooled 95% interval cleared 0 when the analogue's mean gain over the CRF head was about +1.6 or more: at +1.77 the 11 units gave +2.67 [+0.09, +5.25], at +1.27, +2.49 [−0.13, +5.11]. The note added that the decision was the protocol's, whatever these scenarios said; the analogue gained +4.27.

The protocol of 18 October stated that, with the between-subfield standard deviations of weeks 10 and 11 (2.6 and 3.1 points), D1 would pass only for a mean gain of about 2.4–2.9 points or more. The screen's reading as committed at 13:35 on 20 October applied that bar: D2 needed a mean gain near 2.4–2.9, more than the screen's 1.45, while D1's +3.16 was above it. At 18:42 (`de62446`) the reading was redone with V4's own spread: against the CRF head the standard deviation was 5.5 (sofc +12.85, poly −3.72, steel −2.00), so the half-width over 7 subfields was 5.1 and D1 needed a mean near 5, and D2 (standard deviation 3.0, half-width 2.8) a mean near 3, bars that the screen's +3.16 and +1.45 did not reach. The same reading put D5's half-width near 2.9 for a pooled mean near 3.0 ([+0.1, +5.9] with the screen's values), with MR3's spread on the four MaterioMiner papers (0.8) as a proxy.

Week 13. On 24 October, before `w13a` was pushed, the stored T4 runs showed SOFC-Exp's article folds costing more than the protocol's 4.5 and 7.5 lane hours: the CRF head had taken 2.9 hours for its 10 seeds on the 5 folds, and on the official split MR3 took 3.1 times the CRF head's run time and multi-task MR3 4.3 times, which the note put at about 8.8 and 12.5 lane hours. One session of both lanes would have put the multi-task lane past the runner's 10-hour deadline, after which no job starts and a running job is stopped 1.25 hours later. `w13a` was therefore split: it ran MR3 over both lanes, and a new batch, `w13c`, ran multi-task MR3, one after the other on `ner-gpu-jobs-c`, with the runs, their files and the analysis unchanged. The split was committed at 12:23:09 (`d6db00b`), in the minute `w13a` was pushed (version 5); its note is headed 12:30. `w13a` took 293.4 minutes and `w13c` 549.1, its multi-task jobs 77 to 120 minutes each (mean 103) instead of the 75 estimated. In `w13b` (`ner-gpu-jobs-d`, version 13; pushed on 25 October at 01:01, done at 11:21 after 619.8 minutes), in both lanes every fold's three teachers equalled the stored week-8 CRF-head runs of their seeds, and the two lanes harvested the same silver mentions per fold (21,376–48,943, 4,835–9,290 distinct, 20.3–33.4% of the fold's unseen test names). Lane 0's job took 615.2 minutes and lane 1's 591.6, inside the 11.25-hour stop, so no fold was cut; the amendment of 13:53 (13.4, D6's) had costed the batch at about 7.5 hours and the push note at about 8.8.

Week 14. The protocol of 29 October 12:07 (`2886267`) ran `w14b` after the reset of 31 October. With 6.5 of the week's 45 hours unused after `w14a` and both notebooks idle, `w14b` was pushed on 29 October at 18:00 (`ner-gpu-jobs-d`, version 14) from the bundle of `8919928`, the notes recording a change of schedule only. It ran from 18:00 to 22:20 (4.3 session hours). In `w14b` every lane's teachers equalled week 10's on their subfield, and every run carried its mixed corpus's SHA-256 and the mention filter. In `w14c` (31 October, 00:14 to about 09:55, 9.7 session hours) each variant's teachers equalled week 10's on their subfield, and every run carried its topic corpus's SHA-256, the mention filter and the setting it changed. The three new options defaulted to the earlier behavior byte for byte on the tiny model.

The protocol listed what it would not run, and why: V4 on the multi-task model (MaterioMiner only, the names typed at both levels, no second corpus to repeat it on); the copy control on SOFC-Exp's article folds (about 13 lane hours on a corpus where the method does not pass); the names as a match feature (screened in week 12 as V7 and V8, below mention replacement); and the names as a dictionary. The names on the multi-task model were run after the write-up of D1 to D8, in the follow-up of 31 October (19:15) (D9, D10). MR6 was not run on SOFC-Exp's folds.

For a failing D8 the protocol fixed this reading: "A fail with the mixed variant above the CRF head (reported) means a large materials corpus of other topics does as well, a simpler method the paper then states, with 13.2 [D5] as its decision on the topic corpora and this as the reported comparison; a fail with neither above the other means 7 units cannot separate them, which the paper says". At 12:32 (`102144c`), before any run of `w14b`, the protocol added the mixed corpora's coverage of the unseen test names (49–70%, against the topic corpora's 55–81%) and the sentence "So the mixed corpora hold most of what the topic corpora hold, which is what 14.2 [D8] tests: whether the names the topic adds, and the contexts they come in, are what the gain needs". D8 failed, −0.18 [−2.01, +1.65], with the mixed variant at +3.38 [−0.89, +7.64] against the CRF head, and the outcome note of 22:38 read it by the first reading: "the topic's text is not what the gain needs; a large unlabeled materials corpus of other topics gives the same names-and-copies gain". A narrower framing, citing the protocol's sentence of 12:32, entered an earlier draft on 31 October as one added after the result; the main text (Results, *The names' source and the lexicon's settings*) reads the fail by both: the first reading's condition holds on the point estimate but not on the interval, "does as well" being an equivalence D8 cannot establish, and the amendment of 12:32, made before any of D8's runs, had already framed D8 as testing whether the names the topic adds are what the gain needs.

Week 17. `smokebd`, a repeat of week 9's stored seed-0 `crf_mt_mr` run on paper s11340, was identical on both heads (54.05 fine-grained, 62.29 coarse-grained, 887 augmented sentences, the same predictions), so the week-17 code had left the plain multi-task and mention-replacement paths as they were. Every names job's three teachers equalled the stored `crf_mt` runs' test F1, validation F1 and best epoch on both heads. Two fixes were made before any week-17 number was read: `collect --merge` refused the files because each session's copy carried the means

of its own runs (`test_f1_mean` and the like), which are now recomputed with the summary; and the `week17` analysis looked for the multi-task reference's seed-0 file under a tag it does not carry (`crf_mt.json`), so its section came out empty on the first run, and it now uses `seed_family`, as `ner_mentions` does. The merge then united the two sessions' student files, four papers per file, leaving the committed `w17b` runs unchanged. The note of 1 November 00:30 first gave 49–65% as the single-task lexicons' share of a paper's unseen test names; that was the corpus's coverage, and the note was corrected on 1 November to 8.4–18.6% fine-grained and 15.5–23.3% coarse-grained.

The odds stated in the methods plan. The plan of 19 October 17:08 (`09ac262`) put D1 on MuLMS at about 0.45, D2 at about 0.3, the MaterioMiner analogue given a MuLMS pass at about 0.6 and a methods paper at about 0.35–0.4. On 20 October at 08:13 (`25690bc`), after the CPU probe and the screen's first half and before the criterion was revised at 09:14, it put D1 for one of V1, V2, V4 and V6 at about 0.3, D2 at about 0.15 and a methods paper at about 0.2. The revision of 09:14 (`808ff4b`) changed no odds. At 13:35 (`5f844bd`), with V4 the screen's winner, the plan put D1 at about 0.5, D2 at about 0.25, the analogue given D1 at about 0.5 and a methods paper through the route at about 0.25, and at 13:53 (`5f7f827`) it added a pass on SOFC-Exp's folds given D1 at about 0.3. At 18:42 (`de62446`) those figures were replaced: D1 about 0.3, D2 about 0.15, the analogue against the CRF head about 0.5, D5 about 0.4, a methods paper through the route about 0.3, and SOFC-Exp's folds about 0.3 given a pass of D1 or D5. The current text keeps the figures of 19 October, of 08:13 and of 18:42; those of 13:35 and 13:53 survive only in the history. On 24 October at 18:15 (`89b7261`), after D1 and D2 had failed, the plan put D5 at about 0.5, passing if the analogue gained about +2 or more, and a methods paper through the names method at about 0.35. After D5 passed (25 October 01:11, `3432988`) it put a pass on SOFC-Exp's folds at about 0.3, and after D7 failed (29 October 17:38, `8919928`) a pass of D8 at about 0.3.

Sessions that failed, hung, lost jobs or were pushed again (weeks 10 to 18).

1. 20 October, `w12a`, `ner-gpu-jobs-d` version 7 (pushed 02:35, bundle `47e2310`): lost every names job but the first of each variant to the resume check, and ended at 04:53 after 137.6 minutes (pulled at 05:03); its names jobs were pushed again as `w12e`.
2. 20 October, `w12c`, `ner-gpu-jobs-c` version 3 (pushed 02:35, bundle `47e2310`): completed in 59 minutes with 4 of its 14 topic jobs done and 10 failed in 0.01 minutes; pushed again at 04:49 as version 4 from bundle `3ac85a8`, which ran all 15 jobs by 08:06 (196.7 minutes).
3. 20 October, `w12e`, `ner-gpu-jobs-d` version 8 (pushed 05:03, bundle `3ac85a8`): the second push of `w12a`'s names jobs, made before the quota was checked; the user was asked to cancel it, it was not canceled, and it ended at 12:25 (441 minutes), found complete at 13:23 at 46.23 of the week's 45 quota hours.
4. 24 October, `w12b:mrm6`, `ner-gpu-jobs-d` version 9 (pushed 12:22): failed without a trace at 13:44 after 82 minutes, about 1.4 GPU hours charged; pushed again unchanged at 13:49 as version 10, which completed at 18:09.
5. 24 October, `w12d:mrm6`, `ner-gpu-jobs-d` version 11 (pushed 18:10): hung before its setup and ended in ERROR at 20:11, uncharged; pushed again at 20:12 as version 12, which completed on 25 October at 00:48.

Every other GPU session of weeks 10 to 18 was pushed once and completed its jobs: on `ner-gpu-jobs-d`, versions 4 (`w10a`), 5 (`w11a`), 6 (`w11b`), 13 (`w13b`), 14 (`w14b`), 15 (`w17a`) and 16 (`w18a`); on `ner-gpu-jobs-c`, versions 5 (`w13a`), 6 (`w13c`), 7 (`w14a`), 8 (`w14c`), 9 (`w15a`) and 10 (`w17b`). The list leaves out the CPU probe, whose container restarted at 02:20, 03:10 and 04:00 on 20 October, killing the graphene probe each time, and the six TPU sessions of 19 October, to none of which a TPU was attached.

Supplementary Note 12: every comparison of the names route

Table S11 lists every comparison of the protocols of 17 October (20:21) to 1 November that the results files hold, 162 rows in all (160 paired comparisons, a gap between unpaired settings and a level), one each: the twelve decisions of Table 2 and the other decisions of those protocols, the screen, every pre-specified comparison the

results files hold but those listed at the end of this Note, and the post hoc ones. Each row gives micro F1 (or the measure the comparison names: recall, as in D11, precision, or F1 on the seen or unseen test names) and, where the results hold them, macro F1 (every type weighted equally), weighted F1 (types weighted by their test entities), strict IOB2 scoring and the votes (the entity-level majority of the three models of each seed triple). Only micro F1 decides, except D11, on recall, week 10's decision on the votes (10.2) and, in the rows of V6's analogue, the companion study's week-5 decision on the votes. The macro and weighted cells of the 11-unit pooled rows are post hoc, recomputed from the run files on 1 November, after the decisions (`ner_robustness.py`, `pooled_measures`, whose micro values equal the rows' to within 0.01). A row's week is the protocol that named the comparison, with the commit time of the protocol or amendment; a post hoc row gives the protocol it re-reads. Strict IOB2 scoring gives the default scoring's numbers in every row that stores both. Macro F1's interval clears 0 where micro F1's does not in 26 rows: 23 per-corpus comparisons, all on MuLMS and every one against the CRF head (13) or MR3 (10), 13 of them on the screen's three seeds, among them D1 (+3.78 [+2.47, +5.10]) and MR3's own decision of 17 October (+2.12 [+1.14, +3.09]); and 3 pooled post hoc rows, the names against MR3 on both seed bases (+1.33 [+0.29, +2.36] on the decision seeds, +1.73 [+0.84, +2.61] on all ten) and against MR6 on all ten (+1.12 [+0.21, +2.03]). Over MR3 those pooled gains also count the three extra copies (MR6 over MR3, in macro F1 +0.50 [−0.31, +1.30] and +0.61 [−0.15, +1.37], in weighted F1 +0.36 [−0.11, +0.82] and +0.32 [−0.26, +0.90]), and the screen's seeds lift the all-ten readings (over MR3 on MuLMS, in macro F1 +1.72 [+0.44, +3.01] on all ten against +1.10 [−0.46, +2.65] on the decision seeds, in weighted F1 +1.62 [−0.05, +3.29] against +1.17 [−1.09, +3.44]). Weighted F1's interval clears 0 where micro F1's does not in 20 rows (16 per-corpus on MuLMS, among them D1 at +4.17 [+0.39, +7.95], 3 on MaterioMiner, and the pooled names against MR3 on all ten); the converse, micro F1 above 0 and macro F1 not, occurs in 8 rows, 7 on MaterioMiner or SOFC-Exp, among them the names against MR3 on MaterioMiner's coarse-grained papers (micro +0.49 [+0.06, +0.92], macro −0.39 [−2.14, +1.36], weighted +0.49 [−0.03, +1.02]). Table S12 gives each row's per-unit values and its votes' macro and weighted F1. Some numbers quoted in the record are not in the tables: mention replacement's pooled gain with MaterioMiner's coarse-grained papers, +1.80 [+0.39, +3.22], which the protocol of 20 October (09:14) records (Methods, *Evaluation protocol*); the same protocol's +2.34 [+0.94, +3.75] with the fine-grained papers, which row 75 gives as +2.35 from per-unit values rounded differently; the reading of 20 October (18:42), which put D5 at about [+0.1, +5.9] from the screen's values; and the two scenarios of 24 October with an assumed MaterioMiner gain, +2.67 [+0.09, +5.25] and +2.49 [−0.13, +5.11] (Supplementary Note 11). The tables leave out the per-type analyses (Supplementary Note 6), the lexicon split (Supplementary Note 5), the multi-task names' recall and precision on the unseen test names against their control (pre-specified on 31 October, 19:15; Supplementary Note 8 and Table S13) and the companion study's own comparisons except the two the names route reports again (label propagation and self-training on MaterioMiner's papers). The protocol of 20 October (09:14) also listed the label-propagation rule's votes as reported; they were never computed. Generated by `names_route_record.py` (`docs/tables/table13_names_route.md`).

Table S11. Every comparison of the names route: week and protocol time, units, comparison, status (a decision with its verdict, the screen, pre-specified, post hoc, or a comparison of an earlier protocol reported again or recomputed), and the mean paired difference with its 95% CI (and, for micro F1 and the votes, the units above 0) in micro F1 (or the measure the comparison names), macro, weighted and strict F1 and the votes; the macro and weighted cells of the 11-unit pooled rows are post hoc. An empty cell: the results do not hold that quantity, or, for the macro and weighted cells of the other pooled rows (70 to 73), it was not computed. A status such as 10.1 names a decision outside Table 2 by its week and number (10.1 is week 10's decision 1; 13.1a as its protocol names it; 13.1d names two decisions, one per granularity), and "pre-specified (13.3)" a comparison pre-specified with that decision.

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
1	10 (17 October 20:21)	MuLMS, 7 subfields (seeds 0–900)	MR3 − CRF head	decision 10.1: no; votes: decision 10.2: passes	+2.10 [−0.31, +4.51], 6/7	+2.12 [+1.14, +3.09]	+2.68 [+0.32, +5.04]	+2.10 [−0.31, +4.51]	+2.71 [+0.22, +5.20], 6/7
2	10 (17 October 20:21)	MuLMS, 5 random pool splits against the 7 subfields (seeds 0–900; not paired)	CRF head on the random pool splits − CRF head on the held-out subfields	pre-specified	+17.06	+15.06	+18.74	+17.06	
3	10 (17 October 20:21)	MuLMS, 7 subfields (seeds 0–900)	MR3 − CRF head: F1 on the unseen test names	pre-specified	+2.19 [−0.25, +4.63]				+2.92 [+0.41, +5.44]
4	10 (17 October 20:21)	MuLMS, 7 subfields (seeds 0–900)	MR3 − CRF head: F1 on the seen test names	pre-specified	+1.30 [−1.21, +3.81]				+1.17 [−1.69, +4.04]
5	11 (18 October 01:13)	MuLMS, 7 subfields (seeds 0–900)	names from the other subfields' sentences − CRF head	decision 11.1: no	+2.04 [−0.85, +4.92], 5/7	+2.37 [+1.27, +3.47]	+2.62 [+0.02, +5.22]	+2.04 [−0.85, +4.92]	+2.48 [−0.44, +5.40], 5/7
6	11 (18 October 01:13)	MuLMS, 7 subfields (seeds 0–900)	names from the other subfields' sentences − MR3	decision 11.2: no	−0.06 [−1.09, +0.96], 3/7	+0.25 [−0.52, +1.02]	−0.06 [−1.19, +1.06]	−0.06 [−1.09, +0.96]	−0.23 [−1.15, +0.68], 3/7
7	11 (18 October 01:13)	MaterioMiner fine, 4 papers	names from the fatigue corpus's sentences − MR3	decision 11.3: no	+0.23 [−0.69, +1.14], 3/4	+0.58 [−1.65, +2.81]	+0.34 [−0.62, +1.30]	+0.23 [−0.69, +1.14]	+0.44 [−1.85, +2.73], 2/4
8	11 (18 October 01:13)	MaterioMiner coarse, 4 papers	names from the fatigue corpus's sentences − MR3	decision 11.4: no	+0.61 [−0.96, +2.18], 3/4	−0.24 [−2.47, +1.98]	+0.66 [−1.28, +2.61]	+0.61 [−0.96, +2.18]	+0.24 [−1.55, +2.02], 2/4
9	11 (18 October 01:13)	MaterioMiner fine, 4 papers	names from the fatigue corpus's sentences − CRF head	pre-specified	+3.00 [+1.98, +4.02], 4/4	+2.92 [−0.80, +6.64]	+3.96 [+1.72, +6.21]	+3.00 [+1.98, +4.02]	+4.05 [+3.03, +5.07], 4/4
10	11 (18 October 01:13)	MaterioMiner coarse, 4 papers	names from the fatigue corpus's sentences − CRF head	pre-specified	+1.89 [−0.31, +4.10], 4/4	+2.55 [−1.00, +6.10]	+2.52 [−0.06, +5.10]	+1.89 [−0.31, +4.10]	+2.50 [+0.48, +4.51], 4/4
11	11 (18 October 01:13)	MaterioMiner fine, 4 papers	the votes: names from the fatigue corpus's sentences − multi-task	pre-specified	+0.40 [−2.76, +3.55], 3/4	−0.68 [−7.69, +6.34]	+1.05 [−1.54, +3.64]	+0.40 [−2.76, +3.55]	
12	11 (18 October 01:13)	MaterioMiner coarse, 4 papers	the votes: names from the fatigue corpus's sentences − multi-task	pre-specified	+2.81 [+0.95, +4.67], 4/4	+5.40 [+1.47, +9.33]	+3.62 [+1.10, +6.14]	+2.81 [+0.95, +4.67]	
13	11 (18 October 01:13)	MuLMS, 7 subfields (seeds 0–900)	names from the other subfields' sentences − CRF head: F1 on the unseen test names	pre-specified	+2.13 [−0.84, +5.09]				+2.58 [−0.39, +5.56]
14	11 (18 October 01:13)	MuLMS, 7 subfields (seeds 0–900)	names from the other subfields' sentences − CRF head: F1 on the seen test names	pre-specified	+1.21 [−0.97, +3.39]				+1.49 [−0.90, +3.88]
15	11 (18 October 01:13)	MaterioMiner fine, 4 papers	names from the fatigue corpus's sentences − CRF head: F1 on the unseen test names	pre-specified	+3.19 [+2.03, +4.35]				+4.26 [+2.31, +6.22]
16	11 (18 October 01:13)	MaterioMiner fine, 4 papers	names from the fatigue corpus's sentences − CRF head: F1 on the seen test names	pre-specified	+2.66 [+0.03, +5.29]				+3.05 [−1.56, +7.65]
17	11 (18 October 01:13)	MaterioMiner coarse, 4 papers	names from the fatigue corpus's sentences − CRF head: F1 on the unseen test names	pre-specified	+2.21 [−0.28, +4.70]				+2.79 [+0.35, +5.24]
18	11 (18 October 01:13)	MaterioMiner coarse, 4 papers	names from the fatigue corpus's sentences − CRF head: F1 on the seen test names	pre-specified	+0.63 [−1.49, +2.75]				+1.54 [−0.58, +3.67]

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
19	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, sentence filter (V1) − MR3	screen, its statistic (no decision)	+0.80 [−0.91, +2.51], 5/7	+1.09 [−1.88, +4.06]	+1.08 [−0.90, +3.07]	+0.80 [−0.91, +2.51]	+0.10 [−1.68, +1.88], 3/7
20	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, sentence filter (V1) − CRF head	screen (no decision)	+2.50 [−2.09, +7.09], 5/7	+1.87 [−2.69, +6.43]	+3.03 [−2.30, +8.36]	+2.50 [−2.09, +7.09]	+2.91 [−2.04, +7.86], 5/7
21	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, mention filter (V2) − MR3	screen, its statistic (no decision)	+0.97 [−0.91, +2.84], 4/7	+1.85 [+0.76, +2.94]	+1.52 [+0.20, +2.84]	+0.97 [−0.91, +2.84]	+0.17 [−1.51, +1.86], 3/7
22	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, mention filter (V2) − CRF head	screen (no decision)	+2.67 [−1.25, +6.59], 5/7	+2.63 [+0.59, +4.67]	+3.47 [−0.22, +7.16]	+2.67 [−1.25, +6.59]	+2.98 [−1.06, +7.02], 6/7
23	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–200)	MR6 (V3) − MR3	screen, its statistic (no decision)	−0.20 [−3.20, +2.80], 5/7	+1.16 [−1.28, +3.61]	+0.14 [−3.21, +3.48]	−0.20 [−3.20, +2.80]	−0.96 [−3.87, +1.95], 4/7
24	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–200)	MR6 (V3) − CRF head	screen (no decision)	+1.50 [−1.65, +4.66], 4/7	+1.95 [−0.26, +4.15]	+2.08 [−0.72, +4.88]	+1.50 [−1.65, +4.66]	+1.85 [−1.68, +5.37], 4/7
25	12 (amendment 1, 19 October 17:15)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names (V4) − MR3	screen, its statistic (no decision); selected	+1.45 [−1.35, +4.25], 5/7	+3.19 [+1.62, +4.76]	+2.67 [+0.75, +4.59]	+1.45 [−1.35, +4.25]	+0.56 [−1.60, +2.72], 5/7
26	12 (amendment 1, 19 October 17:15)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names (V4) − CRF head	screen (no decision)	+3.16 [−1.96, +8.28], 5/7	+3.97 [+0.69, +7.26]	+4.62 [+0.14, +9.09]	+3.16 [−1.96, +8.28]	+3.37 [−1.52, +8.26], 5/7
27	12 (amendment 1, 19 October 17:15)	MuLMS, 7 subfields (seeds 0–200)	self-training on the corpus's silver sentences (V6) − MR3	screen, its statistic (no decision)	+0.16 [−2.31, +2.63], 3/7	−0.65 [−2.59, +1.30]	−1.39 [−4.19, +1.41]	+0.16 [−2.31, +2.63]	−1.11 [−3.74, +1.52], 2/7
28	12 (amendment 1, 19 October 17:15)	MuLMS, 7 subfields (seeds 0–200)	self-training on the corpus's silver sentences (V6) − CRF head	screen (no decision)	+1.87 [+0.20, +3.53], 6/7	+0.14 [−1.86, +2.13]	+0.56 [−1.77, +2.88]	+1.87 [+0.20, +3.53]	+1.70 [−0.44, +3.84], 5/7
29	12 (amendment 2, 19 October 21:03)	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, mention filter (V7) − MR3	screen, its statistic (no decision)	−1.73 [−4.13, +0.67], 2/7	−1.08 [−4.28, +2.12]	−1.88 [−4.87, +1.12]	−1.73 [−4.13, +0.67]	−2.48 [−4.97, +0.01], 1/7
30	12 (amendment 2, 19 October 21:03)	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, mention filter (V7) − CRF head	screen (no decision)	−0.03 [−2.02, +1.97], 3/7	−0.30 [−3.08, +2.48]	+0.07 [−2.43, +2.57]	−0.03 [−2.02, +1.97]	+0.33 [−2.28, +2.94], 3/7
31	12 (amendment 3, 20 October 01:50)	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, sentence filter (V8) − MR3	screen, its statistic (no decision)	−1.85 [−3.59, −0.12], 1/7	−1.39 [−3.45, +0.66]	−2.11 [−4.12, −0.10]	−1.85 [−3.59, −0.12]	−2.49 [−4.27, −0.71], 0/7
32	12 (amendment 3, 20 October 01:50)	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, sentence filter (V8) − CRF head	screen (no decision)	−0.15 [−2.15, +1.86], 3/7	−0.61 [−2.37, +1.14]	−0.16 [−2.63, +2.31]	−0.15 [−2.15, +1.86]	+0.31 [−1.97, +2.60], 5/7
33	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − CRF head	D1: no	+3.19 [−1.13, +7.51], 6/7	+3.78 [+2.47, +5.10]	+4.17 [+0.39, +7.95]	+3.19 [−1.13, +7.51]	+3.23 [−1.15, +7.61], 5/7
34	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR3	D2: no	+0.92 [−1.63, +3.48], 4/7	+1.10 [−0.46, +2.65]	+1.17 [−1.09, +3.44]	+0.92 [−1.63, +3.48]	+0.57 [−2.22, +3.36], 5/7
35	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–900)	MR6 + names − CRF head	pre-specified; D5's MuLMS part	+3.18 [−1.30, +7.66], 5/7	+3.84 [+2.16, +5.52]	+4.30 [+0.47, +8.14]	+3.18 [−1.30, +7.66]	+3.28 [−1.16, +7.71], 6/7
36	12 (18 October 11:00)	MuLMS, 7 subfields (seeds 0–900)	MR6 + names − MR3	pre-specified	+1.08 [−1.12, +3.28], 5/7	+1.72 [+0.44, +3.01]	+1.62 [−0.05, +3.29]	+1.08 [−1.12, +3.28]	+0.56 [−1.58, +2.71], 3/7
37	12 (18 October 11:00)	MaterioMiner fine, 4 papers	MR6 + names − CRF head	D3: passes; D5's MaterioMiner part	+4.27 [+1.39, +7.15], 4/4	+4.07 [+2.08, +6.05]	+5.00 [+1.44, +8.56]	+4.27 [+1.39, +7.15]	+4.99 [+3.27, +6.72], 4/4
38	12 (18 October 11:00)	MaterioMiner coarse, 4 papers	MR6 + names − CRF head	D4: passes	+1.77 [+0.67, +2.88], 4/4	+2.40 [+0.81, +3.99]	+2.35 [+1.03, +3.67]	+1.77 [+0.67, +2.88]	+2.14 [+0.23, +4.05], 4/4
39	12 (the analogue batches, 20 October 00:51)	MaterioMiner fine, 4 papers	MR6 + names − MR3	pre-specified	+1.50 [−1.76, +4.75], 3/4	+1.73 [−0.47, +3.94]	+1.38 [−2.32, +5.07]	+1.50 [−1.76, +4.75]	+1.38 [−1.73, +4.49], 3/4

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
40	12 (the analogue batches, 20 October 00:51)	MaterioMiner coarse, 4 papers	MR6 + names − MR3	pre-specified	+0.49 [+0.06, +0.92], 4/4	−0.39 [−2.14, +1.36]	+0.49 [−0.03, +1.02]	+0.49 [+0.06, +0.92]	−0.12 [−1.65, +1.42], 2/4
41	12 (the analogue batches, 20 October 00:51)	MaterioMiner fine, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − CRF head	reported again; the companion study's week-5 decision (15 October): no; votes: no	+1.74 [−1.16, +4.64], 3/4	+1.90 [−1.11, +4.90]	+1.07 [−1.38, +3.52]	+1.74 [−1.16, +4.64]	+1.12 [−2.71, +4.96], 3/4
42	12 (the analogue batches, 20 October 00:51)	MaterioMiner coarse, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − CRF head	reported again; the companion study's week-5 decision (15 October): no; votes: no	+1.56 [−0.88, +4.00], 3/4	+1.55 [−0.79, +3.89]	+1.57 [−0.90, +4.04]	+1.56 [−0.88, +4.00]	+0.93 [−1.66, +3.53], 3/4
43	12 (the analogue batches, 20 October 00:51)	MaterioMiner fine, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − MR3	post hoc	−1.03 [−4.47, +2.41], 2/4	−0.44 [−1.66, +0.78]	−2.55 [−4.65, −0.46]	−1.03 [−4.47, +2.41]	−2.49 [−5.74, +0.76], 0/4
44	12 (the analogue batches, 20 October 00:51)	MaterioMiner coarse, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − MR3	post hoc	+0.28 [−1.46, +2.01], 2/4	−1.24 [−3.08, +0.60]	−0.29 [−1.39, +0.82]	+0.28 [−1.46, +2.01]	−1.32 [−1.85, −0.80], 0/4
45	13 (20 October 09:14)	MuLMS, 7 subfields (seeds 0–900)	CRF head + label propagation − CRF head	decision 13.1a: passes	+0.82 [+0.18, +1.46], 7/7	+0.31 [+0.18, +0.44]	+0.95 [+0.30, +1.60]	+0.82 [+0.18, +1.46]	
46	13 (20 October 09:14)	MuLMS, 7 subfields (seeds 0–900)	CRF head + label propagation with retyping − CRF head	pre-specified	+1.01 [+0.28, +1.74], 7/7	+0.36 [+0.20, +0.51]	+1.08 [+0.33, +1.83]	+1.01 [+0.28, +1.74]	
47	13 (20 October 09:14)	MuLMS, 7 subfields (seeds 0–900)	MR3 + label propagation − MR3	decision 13.1b: no	+0.57 [−0.12, +1.25], 6/7	+0.23 [+0.10, +0.35]	+0.70 [+0.08, +1.33]	+0.57 [−0.12, +1.25]	
48	13 (20 October 09:14)	MuLMS, 7 subfields (seeds 0–900)	MR3 + label propagation with retyping − MR3	pre-specified	+0.88 [−0.01, +1.76], 7/7	+0.28 [+0.09, +0.46]	+0.91 [+0.11, +1.70]	+0.88 [−0.01, +1.76]	
49	13 (amendment, 20 October 09:25)	MaterioMiner fine, 4 papers	CRF head + label propagation − CRF head	the companion study's week-7 decision (16 October), recomputed: no	+0.30 [−0.35, +0.95], 2/4	+0.13 [−0.14, +0.39]	+0.54 [−0.40, +1.48]	+0.30 [−0.35, +0.95]	
50	13 (amendment, 20 October 09:25)	MaterioMiner fine, 4 papers	CRF head + label propagation with retyping − CRF head	the companion study's week-7 comparison (16 October; reported), recomputed	+0.31 [−0.73, +1.35], 3/4	+0.08 [−1.00, +1.16]	+0.39 [−1.16, +1.94]	+0.31 [−0.73, +1.35]	
51	13 (amendment, 20 October 09:25)	MaterioMiner fine, 4 papers	MR3 + label propagation − MR3	decision 13.1d: no	+0.19 [−0.34, +0.73], 3/4	+0.07 [−0.15, +0.29]	+0.39 [−0.25, +1.04]	+0.19 [−0.34, +0.73]	
52	13 (amendment, 20 October 09:25)	MaterioMiner fine, 4 papers	MR3 + label propagation with retyping − MR3	pre-specified	+0.22 [−0.69, +1.12], 3/4	+0.05 [−1.00, +1.11]	+0.23 [−1.20, +1.66]	+0.22 [−0.69, +1.12]	
53	13 (amendment, 20 October 09:25)	MaterioMiner fine, 4 papers	multi-task + MR3 + label propagation − multi-task + MR3	pre-specified	+0.31 [−0.47, +1.09], 3/4	+0.07 [−0.02, +0.15]	+0.51 [−0.32, +1.33]	+0.31 [−0.47, +1.09]	
54	13 (amendment, 20 October 09:25)	MaterioMiner fine, 4 papers	multi-task + MR3 + label propagation with retyping − multi-task + MR3	pre-specified	+0.18 [−1.11, +1.47], 2/4	−0.18 [−0.99, +0.64]	+0.13 [−1.64, +1.90]	+0.18 [−1.11, +1.47]	
55	13 (amendment, 20 October 09:25)	MaterioMiner coarse, 4 papers	CRF head + label propagation − CRF head	the companion study's week-7 decision (16 October), recomputed: passes	+0.12 [+0.01, +0.24], 4/4	+0.11 [−0.18, +0.40]	+0.18 [−0.08, +0.43]	+0.12 [+0.01, +0.24]	
56	13 (amendment, 20 October 09:25)	MaterioMiner coarse, 4 papers	CRF head + label propagation with retyping − CRF head	the companion study's week-7 comparison (16 October; reported), recomputed	+0.64 [+0.29, +1.00], 4/4	+0.46 [−0.02, +0.95]	+0.70 [+0.28, +1.12]	+0.64 [+0.29, +1.00]	
57	13 (amendment, 20 October 09:25)	MaterioMiner coarse, 4 papers	MR3 + label propagation − MR3	decision 13.1d: no	+0.11 [−0.08, +0.31], 4/4	+0.10 [−0.14, +0.34]	+0.15 [−0.11, +0.41]	+0.11 [−0.08, +0.31]	

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
58	13 (amendment, 20 October 09:25)	MaterioMiner coarse, 4 papers	MR3 + label propagation with retyping − MR3	pre-specified	+0.63 [+0.13, +1.14], 4/4	+0.48 [+0.12, +0.83]	+0.67 [+0.26, +1.08]	+0.63 [+0.13, +1.14]	
59	13 (amendment, 20 October 09:25)	MaterioMiner coarse, 4 papers	multi-task + MR3 + label propagation − multi-task + MR3	pre-specified	+0.10 [−0.05, +0.25], 3/4	+0.08 [−0.06, +0.22]	+0.13 [−0.03, +0.29]	+0.10 [−0.05, +0.25]	
60	13 (amendment, 20 October 09:25)	MaterioMiner coarse, 4 papers	multi-task + MR3 + label propagation with retyping − multi-task + MR3	pre-specified	+0.60 [−0.21, +1.41], 4/4	+0.38 [−0.26, +1.02]	+0.62 [−0.18, +1.43]	+0.60 [−0.21, +1.41]	
61	13 (amendment, 20 October 09:25)	SOFC-Exp, 5 folds	CRF head + label propagation − CRF head	pre-specified (13.1c: reported, not a decision)	−0.44 [−0.72, −0.16], 0/5	−0.50 [−0.92, −0.08]	−0.35 [−0.63, −0.07]	−0.44 [−0.72, −0.16]	
62	13 (amendment, 20 October 09:25)	SOFC-Exp, 5 folds	CRF head + label propagation with retyping − CRF head	pre-specified (13.1c: reported, not a decision)	−0.43 [−0.68, −0.18], 0/5	−0.49 [−0.86, −0.11]	−0.34 [−0.59, −0.09]	−0.43 [−0.68, −0.18]	
63	13 (20 October 09:14)	SOFC-Exp, 5 folds	MR3 − CRF head	decision 13.3a: no	−0.48 [−1.36, +0.39], 1/5	−0.50 [−1.31, +0.31]	−0.46 [−1.27, +0.34]	−0.48 [−1.36, +0.39]	
64	13 (20 October 09:14)	SOFC-Exp, 5 folds	multi-task + MR3 − CRF head	decision 13.3b: no	−0.11 [−0.69, +0.46], 1/5	−0.16 [−0.59, +0.28]	−0.08 [−0.56, +0.40]	−0.11 [−0.69, +0.46]	
65	13 (20 October 09:14)	SOFC-Exp, 5 folds	multi-task + MR3 − MR3	pre-specified	+0.37 [−0.11, +0.84], 5/5	+0.34 [−0.46, +1.14]	+0.38 [−0.14, +0.90]	+0.37 [−0.11, +0.84]	
66	13 (20 October 09:14)	SOFC-Exp, 5 folds, slot level	multi-task + MR3, its slot level (no reference at that level)	pre-specified	74.01 [67.89, 80.13] (its level)	70.71 [64.62, 76.80] (its level)	73.84 [67.95, 79.72] (its level)	74.01 [67.89, 80.13] (its level)	
67	13 (amendment 13.4, 20 October 13:53)	SOFC-Exp, 5 folds	MR6 + names − CRF head	D6: no	−0.04 [−1.21, +1.14], 2/5	−0.17 [−1.14, +0.81]	−0.04 [−1.12, +1.05]	−0.04 [−1.21, +1.14]	
68	13 (amendment 13.4, 20 October 13:53)	SOFC-Exp, 5 folds	MR6 + names − MR3	pre-specified	+0.45 [+0.04, +0.86], 5/5	+0.33 [−0.07, +0.73]	+0.43 [+0.03, +0.82]	+0.45 [+0.04, +0.86]	
69	13 (20 October 09:14)	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 + names − CRF head	D5: passes (neither corpus below 0)	+3.58 [+0.95, +6.21], 9/11	+3.92 [+2.87, +4.98]	+4.56 [+2.24, +6.88]		
70	13 (amendment 13.4, 20 October 13:53)	pooled, 16 units: MuLMS 7 (0–900) + MaterioMiner fine 4 + SOFC-Exp 5	MR6 + names − CRF head	pre-specified (13.4)	+2.45 [+0.50, +4.40], 11/16				
71	13 (20 October 09:14)	pooled, 16 units: MuLMS 7 (0–900) + MaterioMiner fine 4 + SOFC-Exp 5	MR3 − CRF head	pre-specified (13.3)	+1.46 [+0.28, +2.64], 11/16				
72	13 (20 October 09:14)	pooled, 9 units: MaterioMiner fine 4 + SOFC-Exp 5	multi-task + MR3 − CRF head	pre-specified (13.3)	+2.93 [+0.14, +5.72], 5/9				
73	13 (20 October 09:14)	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	names from the other subfields' sentences (MuLMS) and the fatigue corpus's (MaterioMiner) − CRF head	post hoc	+2.39 [+0.71, +4.06], 9/11				
74	13 (20 October 09:14)	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 + names − MR3	post hoc	+1.23 [−0.22, +2.68], 8/11	+1.73 [+0.84, +2.61]	+1.53 [+0.26, +2.81]		
75	13 (20 October 09:14)	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR3 − CRF head	post hoc	+2.35 [+0.94, +3.75], 10/11	+2.19 [+1.25, +3.14]	+3.02 [+1.58, +4.47]		
76	13 (20 October 09:14)	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 + names − CRF head	D5′, D5 on the decision seeds (post hoc reading): above 0	+3.58 [+1.04, +6.13], 10/11	+3.89 [+3.01, +4.76]	+4.47 [+2.17, +6.77]		
77	13 (20 October 09:14)	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 + names − MR3	post hoc	+1.13 [−0.51, +2.76], 7/11	+1.33 [+0.29, +2.36]	+1.25 [−0.29, +2.78]		

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
78	13 (20 October 09:14)	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR3 − CRF head	post hoc	+2.46 [+1.00, +3.91], 11/11	+2.56 [+1.56, +3.55]	+3.22 [+1.88, +4.57]		
79	14 (29 October 12:07)	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 + names − MR6	D7: no (neither corpus below 0)	+0.80 [−0.66, +2.26], 7/11	+0.83 [−0.01, +1.67]	+0.89 [−0.40, +2.18]		
80	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6	D7's per-corpus condition: its interval not below 0	+0.51 [−1.51, +2.53], 4/7	+0.53 [−0.56, +1.62]	+0.84 [−1.04, +2.71]	+0.51 [−1.51, +2.53]	+0.35 [−1.67, +2.36], 3/7
81	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–900)	MR6 + names − MR6	pre-specified	+0.85 [−1.46, +3.17], 3/7	+0.98 [−0.36, +2.32]	+1.35 [−0.82, +3.51]	+0.85 [−1.46, +3.17]	+0.74 [−1.47, +2.95], 3/7
82	14 (29 October 12:07)	MaterioMiner fine, 4 papers	MR6 + names − MR6	D7's per-corpus condition: its interval not below 0	+1.29 [−2.51, +5.10], 3/4	+1.36 [−0.80, +3.51]	+0.99 [−2.20, +4.18]	+1.29 [−2.51, +5.10]	+0.94 [−1.60, +3.49], 3/4
83	14 (29 October 12:07)	MaterioMiner coarse, 4 papers	MR6 + names − MR6	pre-specified	+0.70 [−1.60, +3.00], 3/4	−0.43 [−3.87, +3.00]	+0.26 [−1.74, +2.25]	+0.70 [−1.60, +3.00]	+1.06 [−0.62, +2.74], 3/4
84	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 300–900)	MR6 − CRF head	pre-specified	+2.68 [+0.13, +5.23], 6/7	+3.25 [+2.60, +3.90]	+3.33 [+1.02, +5.65]	+2.68 [+0.13, +5.23]	+2.88 [−0.03, +5.80], 6/7
85	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 300–900)	MR6 − MR3	pre-specified	+0.41 [−0.35, +1.17], 5/7	+0.56 [−0.40, +1.53]	+0.34 [−0.21, +0.88]	+0.41 [−0.35, +1.17]	+0.22 [−0.94, +1.38], 4/7
86	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–900)	MR6 − CRF head	pre-specified	+2.33 [−0.27, +4.92], 6/7	+2.86 [+2.01, +3.71]	+2.96 [+0.65, +5.27]	+2.33 [−0.27, +4.92]	+2.54 [−0.39, +5.47], 6/7
87	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–900)	MR6 − MR3	pre-specified	+0.23 [−0.39, +0.85], 5/7	+0.74 [−0.07, +1.56]	+0.28 [−0.55, +1.10]	+0.23 [−0.39, +0.85]	−0.17 [−0.94, +0.59], 2/7
88	14 (29 October 12:07)	MaterioMiner fine, 4 papers	MR6 − CRF head	pre-specified	+2.98 [+0.09, +5.86], 4/4	+2.71 [−0.80, +6.22]	+4.01 [+1.37, +6.64]	+2.98 [+0.09, +5.86]	+4.05 [+1.84, +6.27], 4/4
89	14 (29 October 12:07)	MaterioMiner fine, 4 papers	MR6 − MR3	pre-specified	+0.20 [−1.47, +1.88], 3/4	+0.37 [−2.17, +2.92]	+0.38 [−1.13, +1.90]	+0.20 [−1.47, +1.88]	+0.44 [−1.94, +2.82], 2/4
90	14 (29 October 12:07)	MaterioMiner coarse, 4 papers	MR6 − CRF head	pre-specified	+1.07 [−1.82, +3.97], 3/4	+2.83 [−0.65, +6.32]	+2.09 [−0.55, +4.74]	+1.07 [−1.82, +3.97]	+1.08 [−1.68, +3.84], 4/4
91	14 (29 October 12:07)	MaterioMiner coarse, 4 papers	MR6 − MR3	pre-specified	−0.21 [−2.71, +2.29], 2/4	+0.04 [−2.14, +2.23]	+0.24 [−2.12, +2.59]	−0.21 [−2.71, +2.29]	−1.18 [−4.39, +2.03], 2/4
92	14 (29 October 12:07)	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 − CRF head	post hoc	+2.56 [+0.94, +4.18], 10/11	+2.81 [+1.87, +3.75]	+3.34 [+1.86, +4.82]		
93	14 (29 October 12:07)	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 + names − MR6	post hoc	+1.02 [−0.56, +2.59], 6/11	+1.12 [+0.21, +2.03]	+1.22 [−0.21, +2.65]		
94	14 (29 October 12:07)	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 − MR3	post hoc	+0.22 [−0.30, +0.74], 8/11	+0.61 [−0.15, +1.37]	+0.32 [−0.26, +0.90]		
95	14 (29 October 12:07)	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 − CRF head	post hoc	+2.79 [+1.20, +4.38], 10/11	+3.06 [+2.15, +3.96]	+3.58 [+2.12, +5.04]		
96	14 (29 October 12:07)	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 − MR3	post hoc	+0.33 [−0.25, +0.92], 8/11	+0.50 [−0.31, +1.30]	+0.36 [−0.11, +0.82]		
97	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names, own-topic corpus − mixed corpus	D8: no	−0.18 [−2.01, +1.65], 2/7	−0.24 [−1.56, +1.08]	−0.15 [−2.06, +1.75]	−0.18 [−2.01, +1.65]	−0.39 [−2.64, +1.86], 1/7
98	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 300–900)	the mixed-corpus variant − CRF head	pre-specified	+3.38 [−0.89, +7.64], 7/7	+4.02 [+2.38, +5.66]	+4.32 [+0.18, +8.47]	+3.38 [−0.89, +7.64]	+3.62 [−0.71, +7.95], 7/7
99	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 300–900)	the mixed-corpus variant − MR3	pre-specified	+1.10 [−0.76, +2.97], 5/7	+1.33 [+0.02, +2.65]	+1.33 [−0.74, +3.39]	+1.10 [−0.76, +2.97]	+0.96 [−1.02, +2.93], 5/7

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
100	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 300–900)	the mixed-corpus variant − MR6	pre-specified	+0.70 [−1.19, +2.58], 3/7	+0.77 [−0.61, +2.15]	+0.99 [−1.08, +3.06]	+0.70 [−1.19, +2.58]	+0.74 [−0.98, +2.45], 4/7
101	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 10,000 sentences − MR6 + names	pre-specified	−0.15 [−1.78, +1.48], 4/7	−0.73 [−2.44, +0.98]	−0.76 [−2.29, +0.76]	−0.15 [−1.78, +1.48]	+0.20 [−1.06, +1.46], 5/7
102	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 10,000 sentences − MR3	pre-specified	+1.30 [−1.14, +3.74], 5/7	+2.46 [+0.15, +4.78]	+1.91 [−0.57, +4.38]	+1.30 [−1.14, +3.74]	+0.76 [−1.71, +3.23], 3/7
103	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 10,000 sentences − CRF head	pre-specified	+3.01 [−1.21, +7.23], 5/7	+3.25 [+1.06, +5.44]	+3.85 [−0.17, +7.88]	+3.01 [−1.21, +7.23]	+3.57 [−1.12, +8.26], 6/7
104	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 20,000 sentences − MR6 + names	pre-specified	−0.16 [−2.45, +2.14], 4/7	−0.60 [−1.77, +0.57]	−0.77 [−2.55, +1.02]	−0.16 [−2.45, +2.14]	−0.32 [−2.07, +1.42], 4/7
105	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 20,000 sentences − MR3	pre-specified	+1.30 [−0.56, +3.15], 6/7	+2.59 [+1.10, +4.08]	+1.90 [+0.03, +3.78]	+1.30 [−0.56, +3.15]	+0.24 [−1.90, +2.37], 3/7
106	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 20,000 sentences − CRF head	pre-specified	+3.00 [−0.49, +6.49], 6/7	+3.37 [+0.67, +6.08]	+3.85 [+0.51, +7.19]	+3.00 [−0.49, +6.49]	+3.05 [−1.07, +7.16], 6/7
107	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: marginal threshold 0.7 − MR6 + names	pre-specified	−0.16 [−1.62, +1.31], 4/7	−0.31 [−1.43, +0.81]	−0.18 [−1.90, +1.54]	−0.16 [−1.62, +1.31]	+0.34 [−1.07, +1.75], 4/7
108	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: marginal threshold 0.7 − MR3	pre-specified	+1.29 [−0.95, +3.53], 3/7	+2.88 [+1.42, +4.34]	+2.49 [+0.85, +4.13]	+1.29 [−0.95, +3.53]	+0.90 [−1.16, +2.97], 4/7
109	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: marginal threshold 0.7 − CRF head	pre-specified	+3.00 [−1.85, +7.85], 5/7	+3.66 [+0.35, +6.98]	+4.43 [−0.13, +9.00]	+3.00 [−1.85, +7.85]	+3.71 [−1.38, +8.80], 6/7
110	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: 2 of 3 teachers agree − MR6 + names	pre-specified	+0.11 [−1.58, +1.81], 4/7	−0.39 [−1.95, +1.17]	+0.01 [−1.75, +1.76]	+0.11 [−1.58, +1.81]	+0.45 [−1.07, +1.97], 5/7
111	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: 2 of 3 teachers agree − MR3	pre-specified	+1.56 [+0.07, +3.05], 6/7	+2.80 [+1.45, +4.15]	+2.68 [+1.44, +3.92]	+1.56 [+0.07, +3.05]	+1.01 [−0.87, +2.89], 4/7
112	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: 2 of 3 teachers agree − CRF head	pre-specified	+3.27 [−0.85, +7.38], 5/7	+3.58 [+0.46, +6.71]	+4.62 [+0.87, +8.37]	+3.27 [−0.85, +7.38]	+3.82 [−0.24, +7.88], 5/7
113	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, pool: harvested names in half of the draws − MR6 + names	pre-specified	+0.09 [−1.63, +1.80], 3/7	−0.38 [−2.20, +1.45]	−0.27 [−1.84, +1.31]	+0.09 [−1.63, +1.80]	+0.08 [−1.25, +1.42], 5/7
114	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, pool: harvested names in half of the draws − MR3	pre-specified	+1.54 [−0.74, +3.82], 6/7	+2.81 [+0.70, +4.92]	+2.41 [−0.02, +4.83]	+1.54 [−0.74, +3.82]	+0.64 [−1.92, +3.21], 5/7
115	14 (29 October 12:07)	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, pool: harvested names in half of the draws − CRF head	pre-specified	+3.24 [−0.73, +7.22], 6/7	+3.60 [+0.32, +6.87]	+4.35 [+0.35, +8.35]	+3.24 [−0.73, +7.22]	+3.45 [−1.20, +8.10], 6/7
116	17 (31 October 19:15)	MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task + MR6	D9: no	+0.96 [−0.01, +1.92], 4/4	+1.41 [−1.16, +3.97]	+0.93 [−0.57, +2.44]	+0.96 [−0.01, +1.92]	+0.19 [−2.28, +2.66], 2/4
117	17 (31 October 19:15)	MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task + MR3	pre-specified	+0.76 [−0.15, +1.68], 4/4	+1.96 [−1.83, +5.75]	+0.74 [−0.17, +1.65]	+0.76 [−0.15, +1.68]	+0.55 [−1.64, +2.74], 3/4
118	17 (31 October 19:15)	MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task	pre-specified	+3.77 [+1.34, +6.21], 4/4	+4.18 [+0.90, +7.46]	+3.85 [+2.39, +5.31]	+3.77 [+1.34, +6.21]	+3.41 [+0.76, +6.07], 4/4
119	17 (31 October 19:15)	MaterioMiner fine, 4 papers	multi-task + MR6 + names − CRF head	pre-specified	+7.49 [+5.96, +9.02], 4/4	+7.94 [+3.37, +12.51]	+7.20 [+5.70, +8.70]	+7.49 [+5.96, +9.02]	+7.06 [+6.01, +8.12], 4/4
120	17 (31 October 19:15)	MaterioMiner fine, 4 papers	multi-task + MR6 − multi-task + MR3	pre-specified	−0.19 [−2.05, +1.67], 1/4	+0.55 [−1.93, +3.03]	−0.19 [−2.23, +1.84]	−0.19 [−2.05, +1.67]	+0.36 [−1.78, +2.50], 3/4
121	17 (31 October 19:15)	MaterioMiner fine, 4 papers	multi-task + MR6 − multi-task	pre-specified	+2.82 [−0.00, +5.64], 4/4	+2.77 [−1.53, +7.08]	+2.92 [+0.92, +4.92]	+2.82 [−0.00, +5.64]	+3.22 [+1.23, +5.21], 4/4

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
122	17 (31 October 19:15)	MaterioMiner fine, 4 papers	multi-task + MR6 − CRF head	pre-specified	+6.54 [+4.14, +8.94], 4/4	+6.53 [+3.58, +9.49]	+6.27 [+4.25, +8.29]	+6.54 [+4.14, +8.94]	+6.87 [+3.47, +10.28], 4/4
123	17 (31 October 19:15)	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task + MR6	D10: no	+0.85 [−0.72, +2.42], 3/4	+0.49 [−1.48, +2.46]	+0.78 [−0.65, +2.20]	+0.85 [−0.72, +2.42]	+0.63 [−1.65, +2.91], 3/4
124	17 (31 October 19:15)	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task + MR3	pre-specified	+0.91 [−0.14, +1.97], 3/4	+1.32 [−1.99, +4.63]	+1.07 [+0.09, +2.05]	+0.91 [−0.14, +1.97]	+1.27 [−0.89, +3.42], 3/4
125	17 (31 October 19:15)	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task	pre-specified	+3.21 [+0.46, +5.96], 4/4	+5.53 [+1.63, +9.44]	+3.87 [+1.01, +6.72]	+3.21 [+0.46, +5.96]	+3.23 [−0.32, +6.78], 3/4
126	17 (31 October 19:15)	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − CRF head	pre-specified	+2.65 [−0.09, +5.38], 4/4	+3.82 [−0.07, +7.71]	+3.33 [+0.39, +6.27]	+2.65 [−0.09, +5.38]	+2.92 [−0.61, +6.45], 3/4
127	17 (31 October 19:15)	MaterioMiner coarse, 4 papers	multi-task + MR6 − multi-task + MR3	pre-specified	+0.06 [−0.74, +0.86], 3/4	+0.83 [−1.41, +3.07]	+0.29 [−0.42, +1.01]	+0.06 [−0.74, +0.86]	+0.64 [−0.83, +2.11], 2/4
128	17 (31 October 19:15)	MaterioMiner coarse, 4 papers	multi-task + MR6 − multi-task	pre-specified	+2.36 [+1.15, +3.56], 4/4	+5.04 [+2.55, +7.54]	+3.09 [+1.61, +4.57]	+2.36 [+1.15, +3.56]	+2.61 [+0.83, +4.38], 4/4
129	17 (31 October 19:15)	MaterioMiner coarse, 4 papers	multi-task + MR6 − CRF head	pre-specified	+1.80 [+0.19, +3.40], 4/4	+3.33 [−0.09, +6.75]	+2.55 [+0.58, +4.52]	+1.80 [+0.19, +3.40]	+2.29 [+0.61, +3.97], 4/4
130	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon holds	D11: passes as registered, with the 3 flagged subfields matched at the grid's end	+11.24 [+3.31, +19.18], 7/7				
131	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: micro F1	D12: no	+3.26 [−0.79, +7.31], 5/7				
132	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: precision on all test entities	pre-specified	+5.78 [−1.22, +12.79], 5/7				
133	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: recall on the unseen test names	pre-specified	+1.12 [+0.09, +2.15], 6/7				
134	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: precision on the unseen test names	pre-specified	+6.17 [−1.27, +13.61], 5/7				
135	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: F1 on the unseen test names	pre-specified	+3.54 [−0.58, +7.67], 5/7				
136	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon does not hold	pre-specified	−1.73 [−3.02, −0.44], 1/7				
137	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: the recall difference on the names the lexicon holds minus that on the others	pre-specified	+12.98 [+4.18, +21.77], 6/7				
138	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 at matched recall − MR6 at bias 0: micro F1	pre-specified	−2.75 [−5.77, +0.27], 2/7				
139	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon holds	pre-specified (D11's reading without the flagged subfields)	+6.05 [−3.33, +15.44], 4/4				
140	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: micro F1	pre-specified (D12's reading without the flagged subfields)	−0.03 [−2.78, +2.72], 2/4				
141	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged	MR6 + names − MR6 at matched recall: precision on all test entities	pre-specified	+0.56 [−6.74, +7.85], 2/4				

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
		electrolysis, pemfc, sofc)							
142	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: recall on the unseen test names	pre-specified	+0.33 [−0.35, +1.01], 3/4				
143	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: precision on the unseen test names	pre-specified	+0.56 [−6.84, +7.96], 2/4				
144	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: F1 on the unseen test names	pre-specified	+0.20 [−2.58, +2.98], 2/4				
145	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon does not hold	pre-specified	−0.88 [−2.63, +0.87], 1/4				
146	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: the recall difference on the names the lexicon holds minus that on the others	pre-specified	+6.94 [−3.59, +17.46], 3/4				
147	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 at matched recall − MR6 at bias 0: micro F1	pre-specified	−0.23 [−1.85, +1.40], 2/4				
148	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon holds	post hoc (D11's reading with the flagged at the nearest recall)	+8.63 [+3.35, +13.90], 7/7				
149	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: micro F1	post hoc (D12's reading with the flagged at the nearest recall)	+1.37 [−0.66, +3.41], 5/7				
150	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: precision on all test entities	post hoc	+3.04 [−1.32, +7.41], 5/7				
151	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: recall on the unseen test names	post hoc	+0.25 [−0.08, +0.58], 5/7				
152	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: precision on the unseen test names	post hoc	+3.24 [−1.32, +7.80], 5/7				
153	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: F1 on the unseen test names	post hoc	+1.54 [−0.45, +3.54], 5/7				
154	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon does not hold	post hoc	−2.07 [−3.68, −0.46], 1/7				
155	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: the recall difference on the names the lexicon holds minus that on the others	post hoc	+10.70 [+4.33, +17.07], 6/7				
156	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 at matched recall − MR6 at bias 0: micro F1	post hoc	−0.86 [−1.93, +0.21], 2/7				

#	week (protocol, UTC)	corpus, units (MuLMS seeds)	comparison (system − reference)	status	Δ micro F1 [95% CI], units above 0	Δ macro F1 [95% CI]	Δ weighted F1 [95% CI]	Δ strict IOB2 micro F1 [95% CI]	votes: Δ micro F1 [95% CI], units above 0
157	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at its best bias of the grid: recall on the unseen test names the lexicon holds	post hoc	+6.52 [+2.15, +10.89], 7/7				
158	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at its best bias of the grid: recall on the unseen test names the lexicon does not hold	post hoc	−4.16 [−5.82, −2.50], 0/7				
159	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at bias 0: recall on the unseen test names the lexicon holds	post hoc	+7.29 [+3.27, +11.30], 7/7				
160	18 (1 November 10:51)	MuLMS, 7 subfields (seeds 300–900)	MR6 at bias 0 − MR6 at matched recall: recall on the unseen test names the lexicon holds	post hoc	+3.96 [−1.06, +8.98], 4/7				
161	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at bias 0: recall on the unseen test names the lexicon holds	post hoc	+5.92 [−1.10, +12.94], 4/4				
162	18 (1 November 10:51)	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 at bias 0 − MR6 at matched recall: recall on the unseen test names the lexicon holds	post hoc	+0.14 [−4.89, +5.16], 1/4				

Table S12. Per row of Table S11 (#) whose results hold per-unit values: the per-unit differences of its micro (or named) comparison and of its votes (MuLMS's subfields; MaterioMiner's papers s11340, ncomms7164, s41467 and met10111458; SOFC-Exp's folds), and its votes' macro and weighted F1 differences with their 95% CIs.

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
1	MuLMS, 7 subfields (seeds 0–900)	MR3 − CRF head	electrolysis +1.42, graphene +1.66, pemfc +2.28, poly −0.73, semi +2.52, sofc +7.39, steel +0.15	electrolysis +2.12, graphene +2.12, pemfc +3.18, poly −0.65, semi +3.38, sofc +7.96, steel +0.88	+3.22 [+1.63, +4.80]	+3.50 [+1.12, +5.88]
5	MuLMS, 7 subfields (seeds 0–900)	names from the other subfields' sentences − CRF head	electrolysis +2.64, graphene +0.38, pemfc +1.61, poly −0.28, semi +1.16, sofc +8.74, steel −0.00	electrolysis +3.41, graphene +1.29, pemfc +2.33, poly −0.62, semi +2.26, sofc +8.86, steel −0.18	+3.18 [+1.75, +4.60]	+3.16 [+0.55, +5.76]
6	MuLMS, 7 subfields (seeds 0–900)	names from the other subfields' sentences − MR3	electrolysis +1.21, graphene −1.28, pemfc −0.67, poly +0.45, semi −1.36, sofc +1.35, steel −0.15	electrolysis +1.29, graphene −0.83, pemfc −0.85, poly +0.03, semi −1.12, sofc +0.91, steel −1.06	−0.04 [−0.75, +0.67]	−0.34 [−1.29, +0.60]
7	MaterioMiner fine, 4 papers	names from the fatigue corpus's sentences − MR3	s11340 −0.50, ncomms7164 +0.12, s41467 +0.42, met10111458 +0.87	s11340 −1.25, ncomms7164 +1.55, s41467 −0.27, met10111458 +1.72	+0.86 [−3.22, +4.95]	+0.26 [−2.18, +2.69]
8	MaterioMiner coarse, 4 papers	names from the fatigue corpus's sentences − MR3	s11340 −0.62, ncomms7164 +1.43, s41467 +0.24, met10111458 +1.38	s11340 −1.17, ncomms7164 +1.35, s41467 +0.89, met10111458 −0.12	+0.92 [−1.88, +3.72]	+0.36 [−1.78, +2.50]
9	MaterioMiner fine, 4 papers	names from the fatigue corpus's sentences − CRF head	s11340 +2.97, ncomms7164 +3.36, s41467 +2.11, met10111458 +3.55	s11340 +4.09, ncomms7164 +4.19, s41467 +3.19, met10111458 +4.73	+3.90 [+1.10, +6.71]	+4.39 [+2.06, +6.72]
10	MaterioMiner coarse, 4 papers	names from the fatigue corpus's sentences − CRF head	s11340 +0.99, ncomms7164 +3.53, s41467 +0.53, met10111458 +2.52	s11340 +2.58, ncomms7164 +4.15, s41467 +1.10, met10111458 +2.15	+4.20 [−0.11, +8.50]	+3.20 [+0.39, +6.01]
11	MaterioMiner fine, 4 papers	the votes: names from the fatigue corpus's sentences − multi-task	s11340 −2.30, ncomms7164 +0.36, s41467 +1.15, met10111458 +2.38			
12	MaterioMiner coarse, 4 papers	the votes: names from the fatigue corpus's sentences − multi-task	s11340 +3.52, ncomms7164 +3.93, s41467 +1.32, met10111458 +2.47			

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
19	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, sentence filter (V1) − MR3	electrolysis +0.17, graphene +0.93, pemfc +0.88, poly −1.40, semi +0.75, sofc +4.55, steel −0.30	electrolysis −0.44, graphene −0.30, pemfc +1.33, poly −2.20, semi +0.03, sofc +3.67, steel −1.38	+0.13 [−3.58, +3.84]	+0.20 [−1.94, +2.34]
20	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, sentence filter (V1) − CRF head	electrolysis +1.10, graphene +2.56, pemfc +2.91, poly −5.39, semi +6.82, sofc +9.98, steel −0.46	electrolysis +2.73, graphene +2.79, pemfc +3.31, poly −6.07, semi +6.95, sofc +10.95, steel −0.29	+2.51 [−3.39, +8.41]	+3.43 [−2.22, +9.09]
21	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, mention filter (V2) − MR3	electrolysis −1.42, graphene +0.96, pemfc −0.01, poly +1.85, semi −0.22, sofc +4.94, steel +0.67	electrolysis −2.07, graphene −0.21, pemfc +0.37, poly +0.98, semi −1.13, sofc +3.65, steel −0.37	+1.13 [−0.54, +2.81]	+0.64 [−0.65, +1.92]
22	MuLMS, 7 subfields (seeds 0–200)	MR3 + names, mention filter (V2) − CRF head	electrolysis −0.49, graphene +2.59, pemfc +2.02, poly −2.14, semi +5.85, sofc +10.37, steel +0.51	electrolysis +1.10, graphene +2.88, pemfc +2.35, poly −2.89, semi +5.79, sofc +10.93, steel +0.72	+3.52 [+1.65, +5.38]	+3.87 [+0.22, +7.52]
23	MuLMS, 7 subfields (seeds 0–200)	MR6 (V3) − MR3	electrolysis +2.04, graphene +0.63, pemfc +1.76, poly +0.54, semi −7.35, sofc +1.19, steel −0.21	electrolysis +1.34, graphene −0.98, pemfc +1.74, poly +0.16, semi −7.55, sofc +0.32, steel −1.76	+0.44 [−1.73, +2.61]	−0.72 [−4.00, +2.56]
24	MuLMS, 7 subfields (seeds 0–200)	MR6 (V3) − CRF head	electrolysis +2.97, graphene +2.25, pemfc +3.79, poly −3.45, semi −1.27, sofc +6.61, steel −0.37	electrolysis +4.51, graphene +2.11, pemfc +3.72, poly −3.71, semi −0.63, sofc +7.60, steel −0.67	+2.82 [−0.00, +5.64]	+2.51 [−0.62, +5.64]
25	MuLMS, 7 subfields (seeds 0–200)	MR6 + names (V4) − MR3	electrolysis +1.13, graphene +3.11, pemfc −0.24, poly +0.27, semi +0.31, sofc +7.42, steel −1.84	electrolysis −1.63, graphene +1.50, pemfc +0.18, poly +0.24, semi +0.16, sofc +5.19, steel −1.72	+1.70 [+0.25, +3.15]	+1.58 [+0.16, +3.00]
26	MuLMS, 7 subfields (seeds 0–200)	MR6 + names (V4) − CRF head	electrolysis +2.06, graphene +4.73, pemfc +1.79, poly −3.72, semi +6.39, sofc +12.85, steel −2.00	electrolysis +1.54, graphene +4.59, pemfc +2.16, poly −3.63, semi +7.08, sofc +12.47, steel −0.63	+4.09 [+0.61, +7.56]	+4.82 [+0.55, +9.08]
27	MuLMS, 7 subfields (seeds 0–200)	self-training on the corpus's silver sentences (V6) − MR3	electrolysis +1.57, graphene +1.55, pemfc −1.73, poly +4.96, semi −2.32, sofc −2.07, steel −0.84	electrolysis −0.45, graphene +0.73, pemfc −2.62, poly +4.22, semi −2.99, sofc −3.88, steel −2.78	−1.98 [−4.07, +0.11]	−2.64 [−5.49, +0.20]
28	MuLMS, 7 subfields (seeds 0–200)	self-training on the corpus's silver sentences (V6) − CRF head	electrolysis +2.50, graphene +3.17, pemfc +0.30, poly +0.97, semi +3.76, sofc +3.36, steel −1.00	electrolysis +2.72, graphene +3.82, pemfc −0.64, poly +0.35, semi +3.93, sofc +3.40, steel −1.69	+0.40 [−2.36, +3.17]	+0.59 [−2.18, +3.36]
29	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, mention filter (V7) − MR3	electrolysis +0.73, graphene −3.49, pemfc −3.03, poly +1.34, semi −5.89, sofc −1.75, steel −0.03	electrolysis −1.79, graphene −4.28, pemfc −3.08, poly +0.92, semi −7.24, sofc −1.38, steel −0.51	−2.16 [−5.74, +1.41]	−2.76 [−5.84, +0.33]
30	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, mention filter (V7) − CRF head	electrolysis +1.66, graphene −1.86, pemfc −1.01, poly −2.65, semi +0.19, sofc +3.68, steel −0.19	electrolysis +1.38, graphene −1.19, pemfc −1.10, poly −2.95, semi −0.32, sofc +5.90, steel +0.58	+0.22 [−3.00, +3.44]	+0.47 [−2.60, +3.55]
31	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, sentence filter (V8) − MR3	electrolysis −1.28, graphene −1.43, pemfc −2.63, poly +0.43, semi −5.55, sofc −1.70, steel −0.80	electrolysis −2.88, graphene −2.00, pemfc −1.54, poly −0.10, semi −6.30, sofc −2.89, steel −1.75	−2.27 [−4.83, +0.29]	−2.89 [−4.93, −0.85]
32	MuLMS, 7 subfields (seeds 0–200)	lexicon as a gazetteer feature, sentence filter (V8) − CRF head	electrolysis −0.35, graphene +0.20, pemfc −0.60, poly −3.56, semi +0.52, sofc +3.73, steel −0.96	electrolysis +0.29, graphene +1.09, pemfc +0.44, poly −3.97, semi +0.62, sofc +4.39, steel −0.66	+0.11 [−1.73, +1.96]	+0.34 [−2.32, +3.01]
33	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − CRF head	electrolysis +2.52, graphene +0.47, pemfc +3.43, poly −2.35, semi +6.12, sofc +11.86, steel +0.28	electrolysis +2.00, graphene −0.42, pemfc +4.03, poly −2.38, semi +7.29, sofc +11.30, steel +0.80	+4.62 [+3.42, +5.82]	+4.30 [+0.49, +8.12]

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
34	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR3	electrolysis +0.89, graphene −1.21, pemfc +1.04, poly −3.02, semi +5.12, sofc +3.62, steel +0.00	electrolysis +0.41, graphene −2.05, pemfc +0.25, poly −3.34, semi +5.68, sofc +3.00, steel +0.02	+0.99 [−0.75, +2.72]	+0.67 [−1.96, +3.30]
35	MuLMS, 7 subfields (seeds 0–900)	MR6 + names − CRF head	electrolysis +2.39, graphene +1.75, pemfc +2.94, poly −2.76, semi +6.20, sofc +12.16, steel −0.40	electrolysis +1.85, graphene +1.25, pemfc +3.40, poly −2.80, semi +7.22, sofc +11.69, steel +0.32	+4.44 [+2.63, +6.26]	+4.48 [+0.72, +8.23]
36	MuLMS, 7 subfields (seeds 0–900)	MR6 + names − MR3	electrolysis +0.96, graphene +0.09, pemfc +0.65, poly −2.03, semi +3.68, sofc +4.76, steel −0.55	electrolysis −0.27, graphene −0.87, pemfc +0.22, poly −2.15, semi +3.84, sofc +3.73, steel −0.56	+1.22 [−0.20, +2.65]	+0.98 [−0.77, +2.72]
37	MaterioMiner fine, 4 papers	MR6 + names − CRF head	s11340 +1.99, ncomms7164 +5.67, s41467 +3.64, met10111458 +5.78	s11340 +4.36, ncomms7164 +4.91, s41467 +4.16, met10111458 +6.55	+5.10 [+1.24, +8.95]	+5.61 [+1.83, +9.39]
38	MaterioMiner coarse, 4 papers	MR6 + names − CRF head	s11340 +1.74, ncomms7164 +2.56, s41467 +0.88, met10111458 +1.91	s11340 +2.51, ncomms7164 +3.59, s41467 +0.77, met10111458 +1.70	+2.94 [−0.81, +6.70]	+2.79 [+0.26, +5.32]
39	MaterioMiner fine, 4 papers	MR6 + names − MR3	s11340 −1.49, ncomms7164 +2.43, s41467 +1.95, met10111458 +3.09	s11340 −0.98, ncomms7164 +2.26, s41467 +0.70, met10111458 +3.53	+2.06 [−1.06, +5.17]	+1.48 [−2.10, +5.05]
40	MaterioMiner coarse, 4 papers	MR6 + names − MR3	s11340 +0.13, ncomms7164 +0.47, s41467 +0.59, met10111458 +0.77	s11340 −1.25, ncomms7164 +0.80, s41467 +0.56, met10111458 −0.58	−0.33 [−2.43, +1.76]	−0.05 [−1.79, +1.69]
41	MaterioMiner fine, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − CRF head	s11340 +2.36, ncomms7164 −0.81, s41467 +1.93, met10111458 +3.49	s11340 +2.35, ncomms7164 −2.48, s41467 +2.10, met10111458 +2.52	+1.82 [−1.83, +5.48]	+0.36 [−2.85, +3.56]
42	MaterioMiner coarse, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − CRF head	s11340 +2.65, ncomms7164 +1.79, s41467 −0.67, met10111458 +2.47	s11340 +2.35, ncomms7164 +1.40, s41467 −1.42, met10111458 +1.42	+0.91 [−2.82, +4.64]	+0.97 [−1.79, +3.73]
43	MaterioMiner fine, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − MR3	s11340 −1.11, ncomms7164 −4.04, s41467 +0.24, met10111458 +0.80	s11340 −2.99, ncomms7164 −5.13, s41467 −1.35, met10111458 −0.49	−1.22 [−2.14, −0.29]	−3.78 [−6.22, −1.34]
44	MaterioMiner coarse, 4 papers	self-training on the fatigue corpus's silver sentences (V6's analogue) − MR3	s11340 +1.04, ncomms7164 −0.31, s41467 −0.96, met10111458 +1.33	s11340 −1.41, ncomms7164 −1.40, s41467 −1.63, met10111458 −0.86	−2.36 [−5.21, +0.48]	−1.87 [−2.57, −1.17]
45	MuLMS, 7 subfields (seeds 0–900)	CRF head + label propagation − CRF head	electrolysis +0.54, graphene +0.47, pemfc +1.35, poly +0.07, semi +0.93, sofc +2.07, steel +0.29			
46	MuLMS, 7 subfields (seeds 0–900)	CRF head + label propagation with retyping − CRF head	electrolysis +0.81, graphene +0.50, pemfc +1.69, poly +0.09, semi +1.13, sofc +2.37, steel +0.50			
47	MuLMS, 7 subfields (seeds 0–900)	MR3 + label propagation − MR3	electrolysis +0.35, graphene +0.20, pemfc +0.61, poly +0.02, semi +0.68, sofc +2.13, steel −0.03			
48	MuLMS, 7 subfields (seeds 0–900)	MR3 + label propagation with retyping − MR3	electrolysis +0.46, graphene +0.30, pemfc +1.27, poly +0.03, semi +1.22, sofc +2.74, steel +0.14			
49	MaterioMiner fine, 4 papers	CRF head + label propagation − CRF head	s11340 +0.52, ncomms7164 −0.00, s41467 +0.76, met10111458 −0.09			
50	MaterioMiner fine, 4 papers	CRF head + label propagation with retyping − CRF head	s11340 +0.52, ncomms7164 −0.41, s41467 +1.11, met10111458 +0.03			
51	MaterioMiner fine, 4 papers	MR3 + label propagation − MR3	s11340 +0.52, ncomms7164 +0.17, s41467 +0.35, met10111458 −0.26			
52	MaterioMiner fine, 4 papers	MR3 + label propagation with retyping − MR3	s11340 +0.52, ncomms7164 +0.05, s41467 +0.80, met10111458 −0.50			
53	MaterioMiner fine, 4 papers	multi-task + MR3 + label propagation − multi-task + MR3	s11340 +0.41, ncomms7164 +0.31, s41467 +0.86, met10111458 −0.33			
54	MaterioMiner fine, 4 papers	multi-task + MR3 + label propagation with retyping − multi-task + MR3	s11340 +0.41, ncomms7164 −0.41, s41467 +1.23, met10111458 −0.51			

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
55	MaterioMiner coarse, 4 papers	CRF head + label propagation − CRF head	s11340 +0.07, ncomms7164 +0.15, s41467 +0.06, met10111458 +0.21			
56	MaterioMiner coarse, 4 papers	CRF head + label propagation with retyping − CRF head	s11340 +0.42, ncomms7164 +0.90, s41467 +0.49, met10111458 +0.77			
57	MaterioMiner coarse, 4 papers	MR3 + label propagation − MR3	s11340 +0.01, ncomms7164 +0.14, s41467 +0.03, met10111458 +0.27			
58	MaterioMiner coarse, 4 papers	MR3 + label propagation with retyping − MR3	s11340 +0.30, ncomms7164 +0.89, s41467 +0.92, met10111458 +0.43			
59	MaterioMiner coarse, 4 papers	multi-task + MR3 + label propagation − multi-task + MR3	s11340 +0.09, ncomms7164 +0.19, s41467 −0.03, met10111458 +0.15			
60	MaterioMiner coarse, 4 papers	multi-task + MR3 + label propagation with retyping − multi-task + MR3	s11340 +0.03, ncomms7164 +1.10, s41467 +0.32, met10111458 +0.94			
61	SOFC-Exp, 5 folds	CRF head + label propagation − CRF head	fold0 −0.75, fold1 −0.30, fold2 −0.41, fold3 −0.17, fold4 −0.57			
62	SOFC-Exp, 5 folds	CRF head + label propagation with retyping − CRF head	fold0 −0.69, fold1 −0.30, fold2 −0.40, fold3 −0.19, fold4 −0.57			
63	SOFC-Exp, 5 folds	MR3 − CRF head	fold0 −0.39, fold1 −0.42, fold2 −0.81, fold3 −1.36, fold4 +0.56			
64	SOFC-Exp, 5 folds	multi-task + MR3 − CRF head	fold0 −0.12, fold1 −0.07, fold2 −0.65, fold3 −0.34, fold4 +0.61			
65	SOFC-Exp, 5 folds	multi-task + MR3 − MR3	fold0 +0.27, fold1 +0.35, fold2 +0.16, fold3 +1.02, fold4 +0.04			
67	SOFC-Exp, 5 folds	MR6 + names − CRF head	fold0 −0.25, fold1 +0.03, fold2 −0.60, fold3 −0.88, fold4 +1.54			
68	SOFC-Exp, 5 folds	MR6 + names − MR3	fold0 +0.13, fold1 +0.45, fold2 +0.21, fold3 +0.47, fold4 +0.97			
69	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 + names − CRF head	electrolysis +2.39, graphene +1.75, pemfc +2.94, poly −2.76, semi +6.20, sofc +12.16, steel −0.40, s11340 +1.99, ncomms7164 +5.67, s41467 +3.64, met10111458 +5.78			
70	pooled, 16 units: MuLMS 7 (0–900) + MaterioMiner fine 4 + SOFC-Exp 5	MR6 + names − CRF head	electrolysis +2.39, graphene +1.75, pemfc +2.94, poly −2.76, semi +6.20, sofc +12.16, steel −0.40, s11340 +1.99, ncomms7164 +5.67, s41467 +3.64, met10111458 +5.78, fold0 −0.25, fold1 +0.03, fold2 −0.60, fold3 −0.88, fold4 +1.54			
71	pooled, 16 units: MuLMS 7 (0–900) + MaterioMiner fine 4 + SOFC-Exp 5	MR3 − CRF head	electrolysis +1.42, graphene +1.66, pemfc +2.28, poly −0.73, semi +2.52, sofc +7.39, steel +0.15, s11340 +3.48, ncomms7164 +3.24, s41467 +1.69, met10111458 +2.69, fold0 −0.39, fold1 −0.42, fold2 −0.81, fold3 −1.36, fold4 +0.56			
72	pooled, 9 units: MaterioMiner fine 4 + SOFC-Exp 5	multi-task + MR3 − CRF head	s11340 +7.16, ncomms7164 +7.04, s41467 +6.25, met10111458 +6.46, fold0 −0.12, fold1 −0.07, fold2 −0.65, fold3 −0.34, fold4 +0.61			
73	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	names from the other subfields' sentences (MuLMS) and the fatigue corpus's (MaterioMiner) − CRF head	electrolysis +2.64, graphene +0.38, pemfc +1.61, poly −0.28, semi +1.16, sofc +8.74, steel −0.00, s11340 +2.97, ncomms7164 +3.36, s41467 +2.11, met10111458 +3.55			
74	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 + names − MR3	electrolysis +0.96, graphene +0.09, pemfc +0.65, poly −2.03, semi +3.68, sofc +4.76, steel −0.55, s11340 −1.49, ncomms7164 +2.43, s41467 +1.95, met10111458 +3.09			
75	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR3 − CRF head	electrolysis +1.42, graphene +1.66, pemfc +2.28, poly −0.73, semi +2.53, sofc +7.39, steel +0.15, s11340 +3.48, ncomms7164 +3.24, s41467 +1.69, met10111458 +2.69			
76	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 + names − CRF head	electrolysis +2.52, graphene +0.47, pemfc +3.43, poly −2.35, semi +6.12, sofc +11.86, steel +0.28, s11340 +1.99, ncomms7164 +5.67, s41467			

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
			+3.64, met10111458 +5.78			
77	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 + names − MR3	electrolysis +0.89, graphene −1.21, pemfc +1.04, poly −3.02, semi +5.12, sofc +3.62, steel +0.00, s11340 −1.49, ncomms7164 +2.43, s41467 +1.95, met10111458 +3.09			
78	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR3 − CRF head	electrolysis +1.63, graphene +1.68, pemfc +2.40, poly +0.67, semi +1.01, sofc +8.24, steel +0.28, s11340 +3.48, ncomms7164 +3.24, s41467 +1.69, met10111458 +2.69			
79	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 + names − MR6	electrolysis +0.13, graphene −1.24, pemfc +1.16, poly −2.28, semi +3.27, sofc +3.32, steel −0.77, s11340 −2.08, ncomms7164 +1.35, s41467 +3.28, met10111458 +2.63			
80	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6	electrolysis +0.13, graphene −1.24, pemfc +1.16, poly −2.28, semi +3.27, sofc +3.32, steel −0.77	electrolysis −0.30, graphene −1.78, pemfc +1.83, poly −2.27, semi +3.79, sofc +1.72, steel −0.56	+0.92 [−0.21, +2.05]	+0.74 [−1.15, +2.64]
81	MuLMS, 7 subfields (seeds 0–900)	MR6 + names − MR6	electrolysis −0.18, graphene −0.12, pemfc +0.21, poly −1.68, semi +4.59, sofc +4.20, steel −1.03	electrolysis −1.19, graphene −0.36, pemfc +0.70, poly −1.49, semi +5.09, sofc +2.77, steel −0.36	+1.04 [−0.13, +2.20]	+1.26 [−0.86, +3.38]
82	MaterioMiner fine, 4 papers	MR6 + names − MR6	s11340 −2.08, ncomms7164 +1.35, s41467 +3.28, met10111458 +2.63	s11340 −0.91, ncomms7164 +0.13, s41467 +2.06, met10111458 +2.48	+1.04 [−0.43, +2.52]	+0.74 [−1.40, +2.88]
83	MaterioMiner coarse, 4 papers	MR6 + names − MR6	s11340 +2.36, ncomms7164 −1.08, s41467 +0.37, met10111458 +1.16	s11340 +2.23, ncomms7164 −0.08, s41467 +0.48, met10111458 +1.63	−0.36 [−3.20, +2.47]	+0.86 [−1.16, +2.87]
84	MuLMS, 7 subfields (seeds 300–900)	MR6 − CRF head	electrolysis +2.40, graphene +1.71, pemfc +2.28, poly −0.07, semi +2.86, sofc +8.54, steel +1.05	electrolysis +2.30, graphene +1.36, pemfc +2.20, poly −0.11, semi +3.51, sofc +9.58, steel +1.36	+3.70 [+2.78, +4.62]	+3.56 [+1.07, +6.05]
85	MuLMS, 7 subfields (seeds 300–900)	MR6 − MR3	electrolysis +0.77, graphene +0.03, pemfc −0.12, poly −0.74, semi +1.85, sofc +0.30, steel +0.77	electrolysis +0.70, graphene −0.27, pemfc −1.58, poly −1.08, semi +1.89, sofc +1.28, steel +0.59	+0.07 [−1.04, +1.17]	−0.07 [−1.11, +0.97]
86	MuLMS, 7 subfields (seeds 0–900)	MR6 − CRF head	electrolysis +2.57, graphene +1.87, pemfc +2.73, poly −1.08, semi +1.62, sofc +7.96, steel +0.62	electrolysis +3.04, graphene +1.61, pemfc +2.70, poly −1.31, semi +2.13, sofc +8.92, steel +0.68	+3.41 [+2.06, +4.76]	+3.21 [+0.70, +5.73]
87	MuLMS, 7 subfields (seeds 0–900)	MR6 − MR3	electrolysis +1.15, graphene +0.21, pemfc +0.45, poly −0.35, semi −0.91, sofc +0.57, steel +0.47	electrolysis +0.92, graphene −0.50, pemfc −0.48, poly −0.66, semi −1.26, sofc +0.96, steel −0.20	+0.19 [−0.63, +1.01]	−0.29 [−1.15, +0.58]
88	MaterioMiner fine, 4 papers	MR6 − CRF head	s11340 +4.07, ncomms7164 +4.32, s41467 +0.36, met10111458 +3.15	s11340 +5.27, ncomms7164 +4.77, s41467 +2.10, met10111458 +4.07	+4.05 [−0.81, +8.92]	+4.87 [+1.71, +8.02]
89	MaterioMiner fine, 4 papers	MR6 − MR3	s11340 +0.60, ncomms7164 +1.08, s41467 −1.33, met10111458 +0.46	s11340 −0.08, ncomms7164 +2.13, s41467 −1.36, met10111458 +1.06	+1.01 [−1.58, +3.60]	+0.73 [−1.54, +3.00]
90	MaterioMiner coarse, 4 papers	MR6 − CRF head	s11340 −0.62, ncomms7164 +3.65, s41467 +0.52, met10111458 +0.74	s11340 +0.27, ncomms7164 +3.68, s41467 +0.29, met10111458 +0.07	+3.30 [−1.69, +8.30]	+1.93 [−1.15, +5.01]
91	MaterioMiner coarse, 4 papers	MR6 − MR3	s11340 −2.23, ncomms7164 +1.55, s41467 +0.23, met10111458 −0.40	s11340 −3.48, ncomms7164 +0.88, s41467 +0.08, met10111458 −2.20	+0.03 [−3.61, +3.67]	−0.91 [−4.32, +2.50]
92	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 − CRF head	electrolysis +2.57, graphene +1.87, pemfc +2.73, poly −1.08, semi +1.62, sofc +7.96, steel +0.62, s11340 +4.07, ncomms7164 +4.32, s41467 +0.36, met10111458 +3.15			
93	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 + names − MR6	electrolysis −0.18, graphene −0.12, pemfc +0.21, poly −1.68, semi +4.59, sofc +4.20, steel −1.03, s11340 −2.08, ncomms7164 +1.35, s41467 +3.28, met10111458 +2.63			

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
94	pooled, 11 units: MuLMS 7 (0–900) + MaterioMiner fine 4	MR6 − MR3	electrolysis +1.15, graphene +0.21, pemfc +0.45, poly −0.35, semi −0.91, sofc +0.57, steel +0.47, s11340 +0.60, ncomms7164 +1.08, s41467 −1.33, met10111458 +0.46			
95	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 − CRF head	electrolysis +2.40, graphene +1.71, pemfc +2.28, poly −0.07, semi +2.86, sofc +8.54, steel +1.05, s11340 +4.07, ncomms7164 +4.32, s41467 +0.36, met10111458 +3.15			
96	pooled, 11 units: MuLMS 7 (300–900) + MaterioMiner fine 4	MR6 − MR3	electrolysis +0.77, graphene +0.03, pemfc −0.12, poly −0.74, semi +1.85, sofc +0.30, steel +0.77, s11340 +0.60, ncomms7164 +1.08, s41467 −1.33, met10111458 +0.46			
97	MuLMS, 7 subfields (seeds 300–900)	MR6 + names, own-topic corpus − mixed corpus	electrolysis +0.49, graphene −0.26, pemfc −0.63, poly −2.53, semi +3.73, sofc −1.54, steel −0.54	electrolysis −0.33, graphene −0.77, pemfc −0.33, poly −3.09, semi +4.49, sofc −2.46, steel −0.23	−0.24 [−1.99, +1.51]	−0.33 [−2.78, +2.11]
98	MuLMS, 7 subfields (seeds 300–900)	the mixed-corpus variant − CRF head	electrolysis +2.04, graphene +0.73, pemfc +4.06, poly +0.18, semi +2.39, sofc +13.40, steel +0.83	electrolysis +2.33, graphene +0.35, pemfc +4.36, poly +0.72, semi +2.80, sofc +13.76, steel +1.03	+4.86 [+3.14, +6.58]	+4.64 [+0.47, +8.81]
99	MuLMS, 7 subfields (seeds 300–900)	the mixed-corpus variant − MR3	electrolysis +0.40, graphene −0.95, pemfc +1.67, poly −0.49, semi +1.39, sofc +5.16, steel +0.54	electrolysis +0.73, graphene −1.28, pemfc +0.58, poly −0.25, semi +1.19, sofc +5.46, steel +0.26	+1.23 [−0.64, +3.09]	+1.00 [−1.38, +3.39]
100	MuLMS, 7 subfields (seeds 300–900)	the mixed-corpus variant − MR6	electrolysis −0.36, graphene −0.98, pemfc +1.79, poly +0.25, semi −0.46, sofc +4.86, steel −0.22	electrolysis +0.03, graphene −1.01, pemfc +2.16, poly +0.83, semi −0.70, sofc +4.18, steel −0.33	+1.16 [−0.61, +2.94]	+1.08 [−1.02, +3.17]
101	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 10,000 sentences − MR6 + names	electrolysis +0.36, graphene −2.43, pemfc +1.87, poly +1.36, semi −2.27, sofc −1.01, steel +1.07	electrolysis +1.09, graphene −1.54, pemfc +1.06, poly +0.94, semi −2.01, sofc +1.09, steel +0.78	−0.28 [−2.46, +1.90]	−0.28 [−1.86, +1.30]
102	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 10,000 sentences − MR3	electrolysis +1.49, graphene +0.68, pemfc +1.63, poly +1.63, semi −1.96, sofc +6.42, steel −0.78	electrolysis −0.54, graphene −0.04, pemfc +1.24, poly +1.18, semi −1.85, sofc +6.28, steel −0.94	+1.42 [−1.48, +4.33]	+1.30 [−1.04, +3.65]
103	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 10,000 sentences − CRF head	electrolysis +2.42, graphene +2.31, pemfc +3.66, poly −2.36, semi +4.11, sofc +11.84, steel −0.94	electrolysis +2.63, graphene +3.05, pemfc +3.22, poly −2.69, semi +5.07, sofc +13.56, steel +0.15	+3.81 [+1.03, +6.59]	+4.53 [+0.11, +8.96]
104	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 20,000 sentences − MR6 + names	electrolysis +0.90, graphene −2.03, pemfc +2.30, poly +0.81, semi −2.79, sofc −3.19, steel +2.90	electrolysis +0.59, graphene −1.84, pemfc +1.75, poly +0.14, semi −3.40, sofc −1.09, steel +1.59	−0.60 [−1.24, +0.04]	−0.86 [−2.29, +0.58]
105	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 20,000 sentences − MR3	electrolysis +2.04, graphene +1.07, pemfc +2.06, poly +1.08, semi −2.47, sofc +4.24, steel +1.05	electrolysis −1.04, graphene −0.34, pemfc +1.93, poly +0.38, semi −3.24, sofc +4.10, steel −0.13	+1.10 [−0.74, +2.95]	+0.73 [−1.33, +2.79]
106	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, corpus: its first 20,000 sentences − CRF head	electrolysis +2.97, graphene +2.70, pemfc +4.09, poly −2.91, semi +3.60, sofc +9.66, steel +0.89	electrolysis +2.13, graphene +2.75, pemfc +3.91, poly −3.49, semi +3.68, sofc +11.38, steel +0.96	+3.49 [+0.10, +6.88]	+3.96 [+0.13, +7.79]
107	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: marginal threshold 0.7 − MR6 + names	electrolysis +0.62, graphene −3.15, pemfc +0.18, poly −0.47, semi +1.05, sofc −0.96, steel +1.62	electrolysis +2.14, graphene −2.31, pemfc −0.53, poly −0.13, semi +1.79, sofc +0.32, steel +1.12	+0.37 [−0.43, +1.18]	+0.45 [−1.31, +2.21]
108	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: marginal threshold 0.7 − MR3	electrolysis +1.75, graphene −0.05, pemfc −0.06, poly −0.20, semi +1.36, sofc +6.47, steel −0.22	electrolysis +0.51, graphene −0.81, pemfc −0.35, poly +0.11, semi +1.95, sofc +5.51, steel −0.60	+2.07 [+0.44, +3.71]	+2.03 [+0.56, +3.51]

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
109	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: marginal threshold 0.7 − CRF head	electrolysis +2.68, graphene +1.58, pemfc +1.97, poly −4.19, semi +7.44, sofc +11.89, steel −0.38	electrolysis +3.68, graphene +2.28, pemfc +1.63, poly −3.76, semi +8.87, sofc +12.79, steel +0.49	+4.46 [+0.74, +8.18]	+5.27 [+0.56, +9.97]
110	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: 2 of 3 teachers agree − MR6 + names	electrolysis −0.33, graphene −1.96, pemfc +1.00, poly +1.54, semi +1.85, sofc −2.74, steel +1.41	electrolysis +0.24, graphene −1.67, pemfc +0.83, poly +3.02, semi +1.61, sofc −1.45, steel +0.58	+0.10 [−0.60, +0.80]	+0.40 [−1.27, +2.08]
111	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: 2 of 3 teachers agree − MR3	electrolysis +0.80, graphene +1.14, pemfc +0.76, poly +1.81, semi +2.16, sofc +4.69, steel −0.43	electrolysis −1.39, graphene −0.17, pemfc +1.01, poly +3.26, semi +1.77, sofc +3.74, steel −1.14	+1.80 [+0.70, +2.90]	+1.99 [−0.01, +3.98]
112	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, filter: 2 of 3 teachers agree − CRF head	electrolysis +1.73, graphene +2.77, pemfc +2.79, poly −2.18, semi +8.24, sofc +10.11, steel −0.59	electrolysis +1.78, graphene +2.92, pemfc +2.99, poly −0.61, semi +8.69, sofc +11.02, steel −0.05	+4.18 [+0.79, +7.57]	+5.22 [+1.60, +8.84]
113	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, pool: harvested names in half of the draws − MR6 + names	electrolysis −0.97, graphene −0.59, pemfc +1.40, poly +1.45, semi −2.28, sofc −1.31, steel +2.90	electrolysis −0.50, graphene +0.35, pemfc +0.83, poly +0.28, semi −2.77, sofc +0.55, steel +1.84	−0.17 [−2.86, +2.52]	−0.12 [−1.78, +1.53]
114	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, pool: harvested names in half of the draws − MR3	electrolysis +0.16, graphene +2.52, pemfc +1.16, poly +1.72, semi −1.97, sofc +6.11, steel +1.06	electrolysis −2.13, graphene +1.85, pemfc +1.01, poly +0.52, semi −2.61, sofc +5.74, steel +0.12	+1.53 [−2.07, +5.13]	+1.46 [−1.08, +4.01]
115	MuLMS, 7 subfields (seeds 0–200)	MR6 + names, pool: harvested names in half of the draws − CRF head	electrolysis +1.09, graphene +4.15, pemfc +3.19, poly −2.27, semi +4.10, sofc +11.54, steel +0.90	electrolysis +1.04, graphene +4.94, pemfc +2.99, poly −3.35, semi +4.31, sofc +13.02, steel +1.21	+3.91 [−0.73, +8.55]	+4.69 [+0.04, +9.35]
116	MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task + MR6	s11340 +1.14, ncomms7164 +0.16, s41467 +1.62, met10111458 +0.90	s11340 −1.21, ncomms7164 −1.08, s41467 +1.70, met10111458 +1.35	+1.63 [−1.05, +4.31]	+0.54 [−1.80, +2.88]
117	MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task + MR3	s11340 +0.82, ncomms7164 +1.47, s41467 +0.07, met10111458 +0.69	s11340 +0.14, ncomms7164 −0.62, s41467 +0.13, met10111458 +2.55	+1.39 [−2.71, +5.49]	+0.56 [−1.92, +3.05]
118	MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task	s11340 +1.63, ncomms7164 +4.95, s41467 +4.78, met10111458 +3.74	s11340 +0.97, ncomms7164 +3.86, s41467 +4.12, met10111458 +4.70	+4.04 [+0.60, +7.47]	+3.61 [+1.37, +5.86]
119	MaterioMiner fine, 4 papers	multi-task + MR6 + names − CRF head	s11340 +7.98, ncomms7164 +8.52, s41467 +6.32, met10111458 +7.15	s11340 +7.36, ncomms7164 +7.70, s41467 +6.15, met10111458 +7.05	+8.62 [−0.03, +17.26]	+6.96 [+6.17, +7.74]
120	MaterioMiner fine, 4 papers	multi-task + MR6 − multi-task + MR3	s11340 −0.33, ncomms7164 +1.31, s41467 −1.54, met10111458 −0.20	s11340 +1.35, ncomms7164 +0.46, s41467 −1.57, met10111458 +1.20	−0.24 [−3.91, +3.42]	+0.02 [−2.71, +2.75]
121	MaterioMiner fine, 4 papers	multi-task + MR6 − multi-task	s11340 +0.48, ncomms7164 +4.78, s41467 +3.16, met10111458 +2.84	s11340 +2.18, ncomms7164 +4.94, s41467 +2.41, met10111458 +3.35	+2.41 [−1.75, +6.56]	+3.08 [+1.27, +4.88]
122	MaterioMiner fine, 4 papers	multi-task + MR6 − CRF head	s11340 +6.83, ncomms7164 +8.35, s41467 +4.71, met10111458 +6.26	s11340 +8.57, ncomms7164 +8.77, s41467 +4.45, met10111458 +5.70	+6.99 [+0.06, +13.92]	+6.42 [+3.43, +9.40]
123	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task + MR6	s11340 +0.79, ncomms7164 +1.39, s41467 −0.51, met10111458 +1.73	s11340 +0.86, ncomms7164 +1.47, s41467 −1.46, met10111458 +1.64	+0.12 [−2.31, +2.56]	+0.63 [−1.38, +2.63]
124	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task + MR3	s11340 +1.20, ncomms7164 +1.41, s41467 −0.06, met10111458 +1.09	s11340 +2.69, ncomms7164 +1.46, s41467 −0.57, met10111458 +1.48	+1.77 [−4.28, +7.83]	+1.39 [−0.78, +3.57]
125	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task	s11340 +3.32, ncomms7164 +4.26, s41467 +0.73, met10111458 +4.51	s11340 +4.70, ncomms7164 +4.70, s41467 −0.01, met10111458 +3.54	+5.55 [+0.05, +11.05]	+4.11 [+0.33, +7.89]
126	MaterioMiner coarse, 4 papers	multi-task + MR6 + names − CRF head	s11340 +2.18, ncomms7164 +4.69, s41467 +0.59, met10111458 +3.13	s11340 +3.76, ncomms7164 +4.92, s41467 −0.24, met10111458 +3.22	+4.34 [−1.55, +10.24]	+3.69 [−0.16, +7.54]
127	MaterioMiner coarse, 4 papers	multi-task + MR6 − multi-task + MR3	s11340 +0.41, ncomms7164 +0.02, s41467 +0.44, met10111458 −0.64	s11340 +1.83, ncomms7164 −0.02, s41467 +0.89, met10111458 −0.16	+1.65 [−3.57, +6.87]	+0.76 [−0.46, +1.99]

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
128	MaterioMiner coarse, 4 papers	multi-task + MR6 − multi-task	s11340 +2.53, ncomms7164 +2.87, s41467 +1.24, met10111458 +2.78	s11340 +3.84, ncomms7164 +3.23, s41467 +1.45, met10111458 +1.90	+5.42 [+1.16, +9.69]	+3.48 [+1.31, +5.65]
129	MaterioMiner coarse, 4 papers	multi-task + MR6 − CRF head	s11340 +1.40, ncomms7164 +3.30, s41467 +1.10, met10111458 +1.40	s11340 +2.90, ncomms7164 +3.44, s41467 +1.23, met10111458 +1.58	+4.22 [−1.07, +9.51]	+3.06 [+0.81, +5.31]
130	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon holds	electrolysis +13.29, graphene +0.22, pemfc +15.59, poly +1.93, semi +12.40, sofc +25.61, steel +9.67			
131	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: micro F1	electrolysis +7.13, graphene −1.31, pemfc +6.43, poly −1.48, semi +2.20, sofc +9.39, steel +0.47			
132	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: precision on all test entities	electrolysis +10.28, graphene −3.05, pemfc +11.03, poly −2.55, semi +6.90, sofc +16.93, steel +0.93			
133	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: recall on the unseen test names	electrolysis +2.36, graphene +0.46, pemfc +1.34, poly −0.14, semi +0.14, sofc +2.80, steel +0.86			
134	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: precision on the unseen test names	electrolysis +11.30, graphene −3.44, pemfc +11.58, poly −2.30, semi +6.92, sofc +18.08, steel +1.05			
135	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: F1 on the unseen test names	electrolysis +7.52, graphene −1.10, pemfc +6.56, poly −1.37, semi +2.31, sofc +9.93, steel +0.96			
136	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon does not hold	electrolysis −3.78, graphene +0.52, pemfc −2.52, poly −1.07, semi −0.81, sofc −2.29, steel −2.16			
137	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at matched recall: the recall difference on the names the lexicon holds minus that on the others	electrolysis +17.07, graphene −0.30, pemfc +18.11, poly +3.00, semi +13.22, sofc +27.90, steel +11.83			
138	MuLMS, 7 subfields (seeds 300–900)	MR6 at matched recall − MR6 at bias 0: micro F1	electrolysis −7.00, graphene +0.06, pemfc −5.28, poly −0.80, semi +1.07, sofc −6.06, steel −1.24			
139	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon holds	graphene +0.22, poly +1.93, semi +12.40, steel +9.67			
140	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: micro F1	graphene −1.31, poly −1.48, semi +2.20, steel +0.47			
141	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: precision on all test entities	graphene −3.05, poly −2.55, semi +6.90, steel +0.93			
142	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: recall on the unseen test names	graphene +0.46, poly −0.14, semi +0.14, steel +0.86			
143	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: precision on the unseen test names	graphene −3.44, poly −2.30, semi +6.92, steel +1.05			
144	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: F1 on the unseen test names	graphene −1.10, poly −1.37, semi +2.31, steel +0.96			

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
145	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon does not hold	graphene +0.52, poly −1.07, semi −0.81, steel −2.16			
146	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at matched recall: the recall difference on the names the lexicon holds minus that on the others	graphene −0.30, poly +3.00, semi +13.22, steel +11.83			
147	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 at matched recall − MR6 at bias 0: micro F1	graphene +0.06, poly −0.80, semi +1.07, steel −1.24			
148	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon holds	electrolysis +8.76, graphene +0.22, pemfc +11.18, poly +1.93, semi +12.40, sofc +16.23, steel +9.67			
149	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: micro F1	electrolysis +2.30, graphene −1.31, pemfc +3.28, poly −1.48, semi +2.20, sofc +4.15, steel +0.47			
150	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: precision on all test entities	electrolysis +3.75, graphene −3.05, pemfc +6.50, poly −2.55, semi +6.90, sofc +8.82, steel +0.93			
151	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: recall on the unseen test names	electrolysis +0.22, graphene +0.46, pemfc −0.16, poly −0.14, semi +0.14, sofc +0.36, steel +0.86			
152	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: precision on the unseen test names	electrolysis +4.25, graphene −3.44, pemfc +6.73, poly −2.30, semi +6.92, sofc +9.48, steel +1.05			
153	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: F1 on the unseen test names	electrolysis +2.43, graphene −1.10, pemfc +3.22, poly −1.37, semi +2.31, sofc +4.36, steel +0.96			
154	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: recall on the unseen test names the lexicon does not hold	electrolysis −4.57, graphene +0.52, pemfc −3.24, poly −1.07, semi −0.81, sofc −3.18, steel −2.16			
155	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 + names − MR6 at matched recall: the recall difference on the names the lexicon holds minus that on the others	electrolysis +13.33, graphene −0.30, pemfc +14.41, poly +3.00, semi +13.22, sofc +19.41, steel +11.83			
156	MuLMS, 7 subfields (seeds 300–900; the flagged at the grid value of the nearest recall)	MR6 at matched recall − MR6 at bias 0: micro F1	electrolysis −2.17, graphene +0.06, pemfc −2.13, poly −0.80, semi +1.07, sofc −0.82, steel −1.24			
157	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at its best bias of the grid: recall on the unseen test names the lexicon holds	electrolysis +6.40, graphene +0.16, pemfc +6.83, poly +1.15, semi +7.78, sofc +13.94, steel +9.38			
158	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at its best bias of the grid: recall on the unseen test names the lexicon does not hold	electrolysis −4.96, graphene −7.45, pemfc −3.24, poly −3.45, semi −1.72, sofc −3.62, steel −4.70			
159	MuLMS, 7 subfields (seeds 300–900)	MR6 + names − MR6 at bias 0: recall on the unseen test names the lexicon holds	electrolysis +6.40, graphene +0.48, pemfc +6.83, poly +4.64, semi +7.78, sofc +14.09, steel +10.78			

#	corpus, units (MuLMS seeds)	comparison	per unit	votes per unit	votes: Δ macro F1 [95% CI]	votes: Δ weighted F1 [95% CI]
160	MuLMS, 7 subfields (seeds 300–900)	MR6 at bias 0 − MR6 at matched recall: recall on the unseen test names the lexicon holds	electrolysis +6.89, graphene −0.25, pemfc +8.76, poly −2.71, semi +4.62, sofc +11.51, steel −1.11			
161	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 + names − MR6 at bias 0: recall on the unseen test names the lexicon holds	graphene +0.48, poly +4.64, semi +7.78, steel +10.78			
162	MuLMS, 4 subfields (seeds 300–900; without the flagged electrolysis, pemfc, sofc)	MR6 at bias 0 − MR6 at matched recall: recall on the unseen test names the lexicon holds	graphene −0.25, poly −2.71, semi +4.62, steel −1.11			

Supplementary Note 13: further tables

Three analyses the Notes above summarize, in full, and the tests of the decisions. Table S13 gives the unseen-name trade of the main text (Results, *What the names do*) for every system against the one it builds on (post hoc, but for MR3 against the CRF head on MuLMS's unseen and seen test names, pre-specified on 17 October; `ner_mentions.py`, `chain`; of the multi-task names' rows against their control on MaterioMiner, recall and precision on the unseen test names are pre-specified, on 31 October, 19:15, and F1 and the seen names' rows post hoc), Table S14 the lexicon split of the main text (Results, *What the names do*) per held-out unit (post hoc; `ner_lexicon_recall.py`), Table S15 the CPU probe of Supplementary Note 7 per subfield (no decision; `silver_lexicon.py`), and Table S16 the t statistic and P values of every decision (computed for reporting; `decision_tests.py`), with, post hoc, each decision's interval and P value when the units whose training sets share documents are allowed to be correlated (Methods, *Statistics and reproducibility*).

Table S13. The unseen-name trade in full: per setting, each system against the one it builds on, the mean paired difference over the units in F1, recall and precision on the unseen and the seen test names and over all entities, with 95% CIs, on all ten seeds and, on MuLMS, on the decision seeds (300–900).

setting	comparison	seeds	part	Δ F1 [95% CI]	Δ recall [95% CI]	Δ precision [95% CI]
MuLMS, 7 subfields	MR3 − CRF head	all ten	unseen names	+2.19 [−0.25, +4.63]	+3.76 [+0.73, +6.79]	−1.10 [−2.67, +0.47]
MuLMS, 7 subfields	MR3 − CRF head	all ten	seen names	+1.30 [−1.21, +3.81]	+5.06 [+1.90, +8.22]	−3.48 [−5.68, −1.29]
MuLMS, 7 subfields	MR3 − CRF head	all ten	all entities	+2.10 [−0.31, +4.51]	+3.89 [+0.91, +6.87]	−1.40 [−2.92, +0.13]
MuLMS, 7 subfields	MR3 − CRF head	decision seeds (300–900)	unseen names	+2.31 [−0.27, +4.89]	+4.05 [+0.94, +7.16]	−1.16 [−3.83, +1.51]
MuLMS, 7 subfields	MR3 − CRF head	decision seeds (300–900)	seen names	+1.83 [−0.76, +4.41]	+5.57 [+2.28, +8.85]	−2.86 [−5.42, −0.30]
MuLMS, 7 subfields	MR3 − CRF head	decision seeds (300–900)	all entities	+2.27 [−0.25, +4.79]	+4.21 [+1.13, +7.30]	−1.38 [−3.80, +1.04]
MuLMS, 7 subfields	MR6 − MR3	all ten	unseen names	+0.22 [−0.45, +0.88]	+0.63 [−0.20, +1.46]	−0.26 [−1.31, +0.79]
MuLMS, 7 subfields	MR6 − MR3	all ten	seen names	+0.18 [−0.98, +1.33]	+0.79 [−0.79, +2.37]	+0.08 [−1.58, +1.74]
MuLMS, 7 subfields	MR6 − MR3	all ten	all entities	+0.23 [−0.39, +0.85]	+0.68 [−0.09, +1.44]	−0.22 [−1.21, +0.77]
MuLMS, 7 subfields	MR6 − MR3	decision seeds (300–900)	unseen names	+0.43 [−0.32, +1.17]	+0.73 [+0.04, +1.42]	−0.05 [−2.04, +1.93]
MuLMS, 7 subfields	MR6 − MR3	decision seeds (300–900)	seen names	+0.27 [−0.51, +1.04]	+0.52 [−0.61, +1.64]	+0.27 [−0.92, +1.46]
MuLMS, 7 subfields	MR6 − MR3	decision seeds (300–900)	all entities	+0.41 [−0.35, +1.17]	+0.69 [+0.07, +1.30]	+0.01 [−1.85, +1.87]
MuLMS, 7 subfields	MR6 + names − MR6	all ten	unseen names	+0.93 [−1.36, +3.22]	+3.24 [+1.36, +5.13]	−2.23 [−4.84, +0.38]
MuLMS, 7 subfields	MR6 + names − MR6	all ten	seen names	+0.77 [−1.18, +2.72]	+2.43 [−1.04, +5.89]	−0.99 [−2.80, +0.81]
MuLMS, 7 subfields	MR6 + names − MR6	all ten	all entities	+0.85 [−1.46, +3.17]	+3.12 [+1.23, +5.00]	−2.17 [−4.69, +0.35]
MuLMS, 7 subfields	MR6 + names − MR6	decision seeds (300–900)	unseen names	+0.56 [−1.44, +2.56]	+2.56 [+0.76, +4.36]	−2.22 [−5.03, +0.60]
MuLMS, 7 subfields	MR6 + names − MR6	decision seeds (300–900)	seen names	+0.47 [−1.39, +2.33]	+1.89 [−0.70, +4.48]	−0.96 [−3.61, +1.68]
MuLMS, 7 subfields	MR6 + names − MR6	decision seeds (300–900)	all entities	+0.51 [−1.51, +2.53]	+2.48 [+0.67, +4.29]	−2.15 [−4.95, +0.66]
MuLMS, 7 subfields	MR6 + names (mixed corpus) − MR6	decision seeds (300–900)	unseen names	+0.64 [−1.33, +2.62]	+2.47 [−0.09, +5.02]	−1.85 [−3.56, −0.15]
MuLMS, 7 subfields	MR6 + names (mixed corpus) − MR6	decision seeds (300–900)	seen names	+0.99 [−0.70, +2.67]	+2.43 [−0.71, +5.58]	−0.02 [−2.48, +2.45]
MuLMS, 7 subfields	MR6 + names (mixed corpus) − MR6	decision seeds (300–900)	all entities	+0.70 [−1.19, +2.58]	+2.51 [+0.04, +4.98]	−1.68 [−3.35, −0.02]
MaterioMiner fine, 4 papers	MR3 − CRF head	all ten	unseen names	+3.19 [+1.66, +4.72]	+4.51 [+2.27, +6.74]	+0.35 [−2.01, +2.71]
MaterioMiner fine, 4 papers	MR3 − CRF head	all ten	seen names	+1.49 [−1.31, +4.30]	+3.38 [−0.02, +6.77]	−0.49 [−6.98, +6.00]
MaterioMiner fine, 4 papers	MR3 − CRF head	all ten	all entities	+2.77 [+1.51, +4.03]	+4.11 [+2.97, +5.25]	+0.32 [−3.23, +3.87]
MaterioMiner fine, 4 papers	MR6 − MR3	all ten	unseen names	+0.46 [−2.16, +3.09]	+1.25 [−0.88, +3.38]	−0.49 [−4.41, +3.43]
MaterioMiner fine, 4 papers	MR6 − MR3	all ten	seen names	+0.56 [−0.37, +1.49]	+1.21 [−1.26, +3.68]	+0.16 [−0.96, +1.29]
MaterioMiner fine, 4 papers	MR6 − MR3	all ten	all entities	+0.20 [−1.47, +1.88]	+0.93 [+0.09, +1.77]	−0.57 [−3.79, +2.64]
MaterioMiner fine, 4 papers	MR6 + names − MR6	all ten	unseen names	+1.26 [−3.40, +5.91]	+0.86 [−3.38, +5.10]	+1.32 [−6.62, +9.25]
MaterioMiner fine, 4 papers	MR6 + names − MR6	all ten	seen names	+1.03 [−0.77, +2.82]	+0.77 [−1.43, +2.97]	+1.02 [−1.12, +3.16]
MaterioMiner fine, 4 papers	MR6 + names − MR6	all ten	all entities	+1.29 [−2.51, +5.10]	+0.88 [−2.82, +4.58]	+1.56 [−5.05, +8.17]
MaterioMiner fine, 4 papers	multi-task + MR3 − multi-task	all ten	unseen names	+3.09 [+0.99, +5.18]	+3.46 [+1.66, +5.27]	+2.58 [−1.00, +6.17]
MaterioMiner fine, 4 papers	multi-task + MR3 − multi-task	all ten	seen names	+3.00 [−0.85, +6.85]	+5.34 [−4.38, +15.06]	+0.62 [−7.19, +8.44]
MaterioMiner fine, 4 papers	multi-task + MR3 − multi-task	all ten	all entities	+3.01 [+0.42, +5.60]	+3.40 [+0.67, +6.13]	+2.17 [−0.43, +4.78]
MaterioMiner fine, 4 papers	multi-task + MR6 − multi-task + MR3	all ten	unseen names	−0.00 [−1.75, +1.74]	−0.42 [−3.49, +2.64]	+0.14 [−2.17, +2.46]
MaterioMiner fine, 4 papers	multi-task + MR6 − multi-task + MR3	all ten	seen names	−1.13 [−4.03, +1.76]	−0.11 [−4.08, +3.86]	−1.78 [−5.15, +1.60]
MaterioMiner fine, 4 papers	multi-task + MR6 − multi-task + MR3	all ten	all entities	−0.19 [−2.05, +1.67]	−0.33 [−3.43, +2.78]	−0.29 [−1.78, +1.19]

setting	comparison	seeds	part	Δ F1 [95% CI]	Δ recall [95% CI]	Δ precision [95% CI]
MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task + MR6	all ten	unseen names	+0.43 [−1.57, +2.44]	+0.58 [−2.93, +4.09]	+0.82 [−3.39, +5.03]
MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task + MR6	all ten	seen names	+2.43 [+0.40, +4.47]	+0.88 [−3.39, +5.16]	+3.39 [+2.65, +4.12]
MaterioMiner fine, 4 papers	multi-task + MR6 + names − multi-task + MR6	all ten	all entities	+0.96 [−0.01, +1.92]	+0.85 [−2.03, +3.73]	+1.46 [−1.21, +4.13]
MaterioMiner coarse, 4 papers	MR3 − CRF head	all ten	unseen names	+1.58 [−0.02, +3.17]	+1.86 [+0.44, +3.27]	+1.30 [−0.53, +3.12]
MaterioMiner coarse, 4 papers	MR3 − CRF head	all ten	seen names	+0.12 [−0.83, +1.06]	+0.73 [−2.72, +4.18]	−0.31 [−2.60, +1.98]
MaterioMiner coarse, 4 papers	MR3 − CRF head	all ten	all entities	+1.28 [+0.06, +2.51]	+1.55 [+0.48, +2.62]	+1.02 [−0.61, +2.65]
MaterioMiner coarse, 4 papers	MR6 − MR3	all ten	unseen names	−0.64 [−3.44, +2.15]	−0.27 [−1.99, +1.45]	−1.00 [−4.87, +2.87]
MaterioMiner coarse, 4 papers	MR6 − MR3	all ten	seen names	+0.51 [−1.18, +2.21]	+2.98 [+0.65, +5.30]	−1.62 [−2.34, −0.91]
MaterioMiner coarse, 4 papers	MR6 − MR3	all ten	all entities	−0.21 [−2.71, +2.29]	+0.45 [−1.38, +2.27]	−0.82 [−3.97, +2.33]
MaterioMiner coarse, 4 papers	MR6 + names − MR6	all ten	unseen names	+1.37 [−1.75, +4.50]	+1.60 [−1.37, +4.57]	+1.16 [−2.24, +4.57]
MaterioMiner coarse, 4 papers	MR6 + names − MR6	all ten	seen names	−0.48 [−3.19, +2.23]	−3.39 [−7.96, +1.18]	+1.90 [−0.56, +4.36]
MaterioMiner coarse, 4 papers	MR6 + names − MR6	all ten	all entities	+0.70 [−1.60, +3.00]	+0.47 [−1.09, +2.03]	+0.90 [−2.08, +3.88]
MaterioMiner coarse, 4 papers	multi-task + MR3 − multi-task	all ten	unseen names	+2.02 [−0.19, +4.22]	+2.36 [+0.31, +4.42]	+1.68 [−0.68, +4.05]
MaterioMiner coarse, 4 papers	multi-task + MR3 − multi-task	all ten	seen names	+2.11 [−0.02, +4.24]	+4.50 [+0.12, +8.89]	−0.01 [−4.00, +3.97]
MaterioMiner coarse, 4 papers	multi-task + MR3 − multi-task	all ten	all entities	+2.30 [+0.49, +4.10]	+2.73 [+0.90, +4.56]	+1.87 [+0.00, +3.74]
MaterioMiner coarse, 4 papers	multi-task + MR6 − multi-task + MR3	all ten	unseen names	+0.28 [−0.80, +1.37]	+0.65 [−0.73, +2.03]	−0.07 [−1.25, +1.11]
MaterioMiner coarse, 4 papers	multi-task + MR6 − multi-task + MR3	all ten	seen names	−0.56 [−1.03, −0.08]	−0.27 [−1.92, +1.37]	−0.83 [−1.80, +0.14]
MaterioMiner coarse, 4 papers	multi-task + MR6 − multi-task + MR3	all ten	all entities	+0.06 [−0.74, +0.86]	+0.41 [−0.69, +1.52]	−0.28 [−1.26, +0.70]
MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task + MR6	all ten	unseen names	+1.20 [−0.82, +3.22]	+1.45 [−0.69, +3.58]	+0.97 [−1.09, +3.03]
MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task + MR6	all ten	seen names	+0.02 [−1.30, +1.34]	−0.96 [−2.09, +0.16]	+0.83 [−1.27, +2.93]
MaterioMiner coarse, 4 papers	multi-task + MR6 + names − multi-task + MR6	all ten	all entities	+0.85 [−0.72, +2.42]	+0.88 [−0.81, +2.58]	+0.82 [−0.64, +2.27]
SOFC-Exp, 5 folds	MR3 − CRF head	all ten	unseen names	−0.43 [−1.66, +0.79]	+0.08 [−0.63, +0.79]	−0.93 [−2.96, +1.10]
SOFC-Exp, 5 folds	MR3 − CRF head	all ten	seen names	−0.50 [−1.27, +0.28]	−0.45 [−1.98, +1.09]	−0.51 [−1.68, +0.66]
SOFC-Exp, 5 folds	MR3 − CRF head	all ten	all entities	−0.48 [−1.36, +0.39]	−0.22 [−0.99, +0.56]	−0.72 [−2.30, +0.87]

Table S14. The lexicon split per unit: per setting and comparison, each held-out unit's recall gain on the unseen test names its lexicon holds and on the others, and their difference (the paired means are Table S3's).

setting	comparison	unit	gain, names in the lexicon	gain, names not in it	in − out
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − MR6	electrolysis	+6.40	−0.61	+7.01
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − MR6	graphene	+0.48	−1.57	+2.05
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − MR6	pemfc	+6.83	+0.93	+5.90
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − MR6	poly	+4.64	+2.03	+2.61
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − MR6	semi	+7.78	+4.85	+2.93
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − MR6	sofc	+14.09	+1.51	+12.58
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − MR6	steel	+10.78	+0.68	+10.10
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − CRF head	electrolysis	+10.80	+3.04	+7.76
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − CRF head	graphene	+1.18	+2.20	−1.02
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − CRF head	pemfc	+8.23	+7.52	+0.71

setting	comparison	unit	gain, names in the lexicon	gain, names not in it	in − out
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − CRF head	poly	+8.45	+5.19	+3.26
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − CRF head	semi	+14.36	+7.41	+6.95
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − CRF head	sofc	+20.67	+14.30	+6.37
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names − CRF head	steel	+11.01	+4.71	+6.30
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 − CRF head	electrolysis	+4.40	+3.65	+0.75
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 − CRF head	graphene	+0.70	+3.77	−3.07
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 − CRF head	pemfc	+1.40	+6.59	−5.19
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 − CRF head	poly	+3.81	+3.16	+0.65
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 − CRF head	semi	+6.58	+2.56	+4.02
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 − CRF head	sofc	+6.58	+12.79	−6.21
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 − CRF head	steel	+0.23	+4.03	−3.80
MuLMS, 7 held-out subfields (seeds 300–900)	MR3 − CRF head	electrolysis	+3.02	+3.81	−0.79
MuLMS, 7 held-out subfields (seeds 300–900)	MR3 − CRF head	graphene	−0.75	+2.30	−3.05
MuLMS, 7 held-out subfields (seeds 300–900)	MR3 − CRF head	pemfc	+4.01	+6.42	−2.41
MuLMS, 7 held-out subfields (seeds 300–900)	MR3 − CRF head	poly	+2.91	+1.38	+1.53
MuLMS, 7 held-out subfields (seeds 300–900)	MR3 − CRF head	semi	+3.89	+1.40	+2.49
MuLMS, 7 held-out subfields (seeds 300–900)	MR3 − CRF head	sofc	+8.31	+11.43	−3.12
MuLMS, 7 held-out subfields (seeds 300–900)	MR3 − CRF head	steel	+3.52	+2.82	+0.70
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names (mixed corpus) − MR6	electrolysis	+6.17	−2.01	+8.18
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names (mixed corpus) − MR6	graphene	−0.81	−1.25	+0.44
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names (mixed corpus) − MR6	pemfc	+7.21	+1.39	+5.82
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names (mixed corpus) − MR6	poly	+4.53	+2.62	+1.91
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names (mixed corpus) − MR6	semi	+3.43	+0.86	+2.57
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names (mixed corpus) − MR6	sofc	+16.40	+5.76	+10.64
MuLMS, 7 held-out subfields (seeds 300–900)	MR6 + names (mixed corpus) − MR6	steel	+6.20	+1.77	+4.43
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	s11340	+1.25	−3.14	+4.39
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	ncomms7164	+13.48	+0.61	+12.87
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	s41467	−0.45	−0.08	−0.37
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	met10111458	+4.19	+3.78	+0.41
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	s11340	+0.00	+6.86	−6.86
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	ncomms7164	+35.22	+5.64	+29.58
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	s41467	+2.96	+3.42	−0.46
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	met10111458	+12.90	+8.08	+4.82
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	s11340	−1.25	+10.00	−11.25
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	ncomms7164	+21.74	+5.03	+16.71
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	s41467	+3.41	+3.50	−0.09
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	met10111458	+8.71	+4.30	+4.41
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	s11340	+0.00	+8.00	−8.00

setting	comparison	unit	gain, names in the lexicon	gain, names not in it	in − out
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	ncomms7164	+21.30	+1.61	+19.69
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	s41467	+4.09	+3.24	+0.85
MaterioMiner fine-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	met10111458	+9.68	+3.81	+5.87
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	s11340	+1.54	+4.85	−3.31
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	ncomms7164	−0.41	−0.16	−0.25
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	s41467	+0.67	+1.75	−1.08
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − MR6	met10111458	−0.34	+1.20	−1.54
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	s11340	+1.15	+5.84	−4.69
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	ncomms7164	+3.06	+3.48	−0.42
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	s41467	+0.67	+2.55	−1.88
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 + names − CRF head	met10111458	+0.75	+2.63	−1.88
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	s11340	−0.39	+0.99	−1.38
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	ncomms7164	+3.47	+3.64	−0.17
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	s41467	+0.00	+0.80	−0.80
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR6 − CRF head	met10111458	+1.09	+1.43	−0.34
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	s11340	+0.38	+2.97	−2.59
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	ncomms7164	+0.82	+3.12	−2.30
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	s41467	−1.26	+1.19	−2.45
MaterioMiner coarse-grained, 4 held-out papers (10 seeds)	MR3 − CRF head	met10111458	−0.41	+1.99	−2.40

Table S15. The CPU probe per held-out MuLMS subfield (no decision): the share of the unseen test names the topic corpus holds and that the sentence and mention filters admit at threshold 0.9; the CRF head's precision and recall (10 seeds); and the change of its micro F1 when every verbatim match of a lexicon name is tagged where it predicted O, for each 0.9 lexicon and for the gold unseen names themselves, which also override its tags in the last column.

subfield	corpus holds (%)	sentence filter admits (%)	mention filter admits (%)	CRF precision	CRF recall	Δ F1, sentence lexicon	Δ F1, mention lexicon	Δ F1, gold names (fill)	Δ F1, gold names (override)
electrolysis	70.7	12.5	29.9	57.3	54.7	+0.85	−0.77	−6.85	+4.47
graphene	64.0	15.3	28.8	56.8	44.9	−2.86	−5.07	+9.11	+24.32
pemfc	73.9	19.2	29.4	60.5	42.9	+3.16	+3.97	−10.51	−5.91
poly	65.9	8.0	18.3	55.9	48.7	−0.25	−3.31	−3.63	+2.22
semi	69.3	4.6	9.4	54.0	23.5	+1.20	+2.27	+17.29	+22.00
sofc	80.8	25.8	34.4	57.5	35.3	+5.79	+5.91	+15.16	+27.06
steel	55.0	13.3	25.0	64.4	50.9	+0.01	−10.03	−1.82	+4.13
mean	68.5	14.1	25.0	58.1	43.0	+1.13	−1.00	+2.68	+11.18

Table S16. The tests of the decisions (Methods, *Statistics and reproducibility*): per decision, the units, the stored mean paired difference with its 95% CI, the standard deviation (SD) of the per-unit differences, the one-sample Student-t statistic with its degrees of freedom, its one- and two-sided P values, and the exact two-sided sign-flip P value over all 2^n assignments of signs, from the per-unit differences each decision used (rounded to two decimals, so t and P can differ from the unrounded values in the last digit); D5′ is the post hoc reading of D5 on MuLMS's decision seeds and not a decision, so its verdict reads "above 0 (post hoc)", and D11 passes as registered only with the three flagged subfields. Computed on 1 November for reporting: each decision was taken by the registered rule, the interval above 0 (for D5 and D7 with neither corpus's interval below 0), which is the one-sided test at 2.5%. The last column (post hoc) gives the interval and two-sided P value with the units correlated where their training sets share documents, at the correlation n_test/n that the corrected resampled t-test assumes, 1/K in a design of K folds: 1/4 between MaterioMiner's papers, 1/5 between SOFC-Exp's folds, 1/4 between the 5 of 21 pairs of MuLMS subfields whose training pools share an article, none otherwise or across corpora; the variance of the mean of n units is then $S^2/(1 - \bar{\rho})\cdot(n + \Sigma\rho)/n^2$, $\Sigma\rho$ over the $n(n - 1)$ ordered pairs $i \neq j$ and $\bar{\rho} = \Sigma\rho/(n(n - 1))$, with $n - 1$ degrees of freedom (`decision_tests.py`, `post_hoc_dependence`).

Decision or reading	Comparison	Units	Δ [95% CI]	SD	t (df)	P, one-sided	P, two-sided	Sign-flip P	Verdict	Units correlated: 95% CI, P (post hoc)
D1	MuLMS: MR6 + names vs CRF head	7	+3.19 [−1.13, +7.51]	4.67	1.81 (6)	0.0603	0.1207	0.0781	no	[−2.00, +8.38], 0.183
D2	MuLMS: MR6 + names vs MR3	7	+0.92 [−1.63, +3.48]	2.76	0.88 (6)	0.2062	0.4125	0.4062	no	[−2.15, +3.99], 0.491
D3	MaterioMiner fine: MR6 + names vs CRF head	4	+4.27 [+1.39, +7.15]	1.81	4.72 (3)	0.0090	0.0180	0.1250	passes	[−0.13, +8.67], 0.054
D4	MaterioMiner coarse: MR6 + names vs CRF head	4	+1.77 [+0.67, +2.88]	0.69	5.12 (3)	0.0072	0.0144	0.1250	passes	[+0.09, +3.45], 0.044
D5	pooled 11 units (MuLMS seeds 0–900): MR6 + names vs CRF head	11	+3.58 [+0.95, +6.21]	3.92	3.03 (10)	0.0063	0.0127	0.0107	passes	[+0.27, +6.88], 0.037
D5′	pooled 11 units (MuLMS seeds 300–900, post hoc reading): MR6 + names vs CRF head	11	+3.58 [+1.04, +6.13]	3.79	3.14 (10)	0.0053	0.0106	0.0068	above 0 (post hoc)	[+0.38, +6.78], 0.032
D6	SOFC-Exp folds: MR6 + names vs CRF head	5	−0.04 [−1.21, +1.14]	0.94	−0.08 (4)	0.5284	0.9432	1.0000	no	[−1.79, +1.73], 0.962
D7	pooled 11 units: MR6 + names vs MR6	11	+0.80 [−0.66, +2.26]	2.17	1.22 (10)	0.1258	0.2517	0.2383	no	[−1.04, +2.63], 0.356
D8	MuLMS: own-topic vs mixed corpus	7	−0.18 [−2.01, +1.65]	1.98	−0.24 (6)	0.5926	0.8148	0.7969	no	[−2.38, +2.01], 0.845
D9	MaterioMiner fine: multi-task + MR6 + names vs multi-task + MR6	4	+0.96 [−0.01, +1.92]	0.61	3.14 (3)	0.0259	0.0517	0.1250	no	[−0.52, +2.43], 0.132
D10	MaterioMiner coarse: multi-task + MR6 + names vs multi-task + MR6	4	+0.85 [−0.72, +2.42]	0.99	1.72 (3)	0.0916	0.1833	0.2500	no	[−1.55, +3.25], 0.341
D11	MuLMS: MR6 + names vs recall-matched MR6, recall on the unseen names the lexicon holds	7	+11.24 [+3.31, +19.18]	8.58	3.47 (6)	0.0067	0.0133	0.0156	passes (as registered)	[+1.71, +20.77], 0.028
D12	MuLMS: MR6 + names vs recall-matched MR6, micro F1	7	+3.26 [−0.79, +7.31]	4.38	1.97 (6)	0.0480	0.0961	0.1406	no	[−1.60, +8.12], 0.152